METHOD AND APPARATUS FOR VENDOR-FREE DOMAIN-SEPARATED DEVICE ATTESTATION VIA OWNER-INITIATED CRYPTOGRAPHIC PROVISIONING

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1 CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is one of four related filings establishing a complete stack of user-sovereign computing infrastructure. The related applications are: SPIRIX (filed 2025), directed to a two's complement floating-point arithmetic format providing cross-platform deterministic computation; ISOMEM (provisional filed March 2026), directed to hardware memory protection architecture using base-limit register pairs with cryptographically indistinguishable invalid access responses; and TOKEN (to be filed), directed to user-sovereign human identity, threshold reconstitution, billing attestation, mobile virtual network operator network provisioning, and physics-anchored temporal sovereignty. The dependencies among these applications are described in the cross-reference section near the end of this specification.

2 STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

[0002] Not applicable.

3 BACKGROUND OF THE INVENTION

3.1 Field of the Invention

[0003] The present invention relates to hardware-rooted cryptographic device identity, hardware identity element design, single-channel bidirectional communication protocols, line coding, true-random-number generation, write-once cryptographic storage, and domain-separated cryptographic attestation. More specifically, the present invention pertains to a chip architecture wherein the chip's permanent cryptographic identifier originates from physical entropy sampled by noise-capture circuitry resident on the chip's silicon at the unrepeatable moment of an owner-initiated provisioning event, the sampled entropy comprising transient physical phenomena present at and within the chip's silicon at the moment of sampling, the unrepeatability of the entropy deriving jointly from the owner's choice of triggering moment and from the unrepeatable physical state present at that moment; is never transmitted, exposed, or otherwise made accessible to any party outside the chip by any non-destructive means; and functions only as a key for cryptographic commitments produced inside the chip and emitted to requesting applications as domain-separated derivatives carried over a single bidirectional communication channel, the channel comprising any of an electrical conductor, an optical fiber coupled to paired light emitters and detectors, a magnetic or inductive coupling element, a capacitive coupling element, a radio-frequency near-field link, or an acoustic link, such that the chip's identifier has no issuing authority and the chip's external behaviour does not enable cross-application correlation of the device.

[0004] The present invention does not address human identity, user authentication, account systems, or service authorization. Those concerns are the subject of the related TOKEN application identified above and are outside the scope of the present filing.

3.2 Description of the Related Art

[0005] A purchaser of an integrated-circuit hardware identity device does not, in current commercial practice, purchase the device's cryptographic identity. Every hardware identity element commercially distributed in the past has a cryptographic key whose value was known to the chip's manufacturer prior to the purchaser's acquisition of the chip, or whose value is subject to certification by a certificate authority controlled by the chip's manufacturer, or both. Commercial implementations of this pattern include the Apple Secure Enclave, the Google Titan and Titan M security chips, the Microsoft Pluton security processor, the ARM TrustZone CPU operating mode, the Qualcomm QSEE environment, the Intel SGX enclave system, and Trusted Platform Modules conforming to the Trusted Computing Group TPM 2.0 specification. In each such implementation, the chip's manufacturer occupies a position in the trust path during identity creation, attestation signing, certificate authority operation, or destruct command authority. The user, who purchased the chip and holds the chip physically, does not occupy any of these positions in the trust path.

[0006] A second structural problem affects prior art devices independently of the first. Even when the chip's manufacturer were entirely honest, and even when the manufacturer never used or disclosed its knowledge of the chip's key, the chip in prior art implementations answers cryptographic identity queries using a single fingerprint-stable key. Two unrelated services querying the same prior art chip with the same question receive answers that any third party can correlate, by inspection of the answers, without requiring access to either service's secret material. The chip thereby functions as a structural surveillance surface. Cross-service correlation of users does not require a manufacturer's cooperation; it requires only that the operators of multiple services maintain logs of the chip-attested values they have received, and that those logs be joined on the fingerprint field. The FIDO2 specification family partially addresses this property within FIDO authentication flows thru the use of per-relying-party credential key pairs, each registration with a relying party generating a fresh key pair independently of any prior relying party's key pair, such that two relying parties holding

the same authenticator's public keys cannot correlate them by inspection; however, the authenticator's attestation statement to its hardware identity is signed by an attestation key whose root certificate is manufacturer-controlled, returning manufacturer trust to the path. Furthermore, the per-relying-party credential isolation is achieved by separately generating each key pair rather than by cryptographic domain separation of a shared internal secret, and so does not extend to uses of the device's identity outside FIDO authentication flows.

[0007] The pattern is consistent across the prior art. Chip identity, as the prior art implements it, is a permission granted by the chip's manufacturer and revocable by the chip's manufacturer; the chip's user is a tenant in the manufacturer's identity infrastructure rather than the owner of the chip's identity. Furthermore, prior art chip identity answers every cryptographic query with a structurally correlatable value, rendering the chip a single fingerprint regardless of the consent of the chip's user to such correlation.

[0008] Specific prior art systems and their limitations are summarized as follows. The Apple Secure Enclave provides per-chip unique identifiers generated at fabrication and signed by Apple's certificate authority; Apple retains provisioning records, can revoke device attestation, and the single per-chip identifier enables fingerprint correlation across all services that receive Apple-attested device assertions. Google Titan and Titan M security chips provide functionally equivalent vendor-attested keys under Google's certificate authority, with the same structural properties. Trusted Platform Modules conforming to TPM 2.0 provide an endorsement key (EK) generated at manufacture and certified by the TPM vendor; the EK cannot be replaced; the EK certificate identifies the manufacturer and the chip model; the Storage Root Key and Attestation Identity Keys are derived from the EK; and the manufacturer's signature serves as the trust root for all subsequent attestations.

[0009] ARM TrustZone provides a CPU operating mode, not a physically isolated enclave, and therefore shares physical resources (caches, branch predictors, memory buses, and power rails) with the rich execution environment; this property has resulted

in repeated side-channel compromise (including the Foreshadow, Spectre, and ÆPIC vulnerability families). Microsoft Pluton is provisioned by original equipment manufacturers under Microsoft attestation, exhibiting the same vendor trust path structure. Intel SGX enclaves are attested by the Intel Attestation Service; the SGX system has been compromised via side channels and was structurally retired from Intel's consumer chip product lines in 2021, with the consequence that every user who had based identity infrastructure on SGX had that infrastructure revoked by Intel as a corporate decision.

[0010] FIDO2 hardware token products including YubiKey, Solokey, Trezor, and Nitrokey, provide per-device cryptographic keys generated on the device; however, the firmware controlling the device is signed by the device vendor, the attestation root certificate is vendor-controlled, and the per-relying-party key separation operates only within FIDO authentication flows. For any non-FIDO use of the device key (including TPM remote attestation, automotive vehicle identity binding, and IoT device registration), the single-fingerprint property of the underlying device key returns.

[0011] Hardware identity element products including the STMicroelectronics STSAFE family, the NXP A71CH, the Infineon OPTIGA family, and the Microchip ATECC family, provide factory-injected keys and vendor-signed device certificates. The chip ships with the key already provisioned by the manufacturer.

[0012] Single-wire bidirectional communication prior art (including the Dallas / Maxim 1-Wire family and the Philips I^2C bus) provides physical-layer prior art for single-conductor bidirectional signaling but does not include cryptographic protocols, domain separation mechanisms, entropy semantics in the handshake, or silent-on-corruption behaviour at the wire level. Line-coding prior art for direct-current balance and clock recovery (including Manchester encoding, 8b/10b coding, and NRZ-I) requires dedicated overhead bits and explicit encoder and decoder hardware, and does not deliver protocol semantics within the line code itself.

[0013] Cryptographic prior art for authenticated session protocols (including the Signal

Protocol, the Noise Protocol Framework, and WireGuard) provides individual instances of any one of the five protocol functions combined in the present invention's masked payload fields, but delivers those functions thru separate transmitted fields and does not combine them within the masked fields at the wire level. Cryptographic prior art for keyed cryptographic commitment functions (including BLAKE3 keyed hash, HKDF, HMAC, and the Poly1305 authenticator) provides commitment function prior art alone, but does not combine such functions with hardware-isolated permanent identifiers wherein use of the commitment function is enforced at the silicon level rather than the software level.

[0014] Physical Unclonable Function (PUF) prior art (including Verayo Vera M4H, Intrinsic ID SRAM PUF technology, and Maxim PUF-based cryptographic cells) provides hardware-rooted identity derived from manufacturing-process variation. PUF identities in commercial use are typically read out at provisioning time and enrolled in a vendor-controlled database, the stated purpose of the enrolment being to enable subsequent vendor certification of the chip's identity; the structural effect of such enrolment, irrespective of stated purpose, is that the vendor retains a queryable record of the chip's identity and therefore remains in the trust path. SRAM PUF embodiments operated without enrolment retain the single-fingerprint property and do not provide domain separation.

[0015] Trusted Computing Group Direct Anonymous Attestation (DAA) provides a cryptographic scheme allowing a Trusted Platform Module to prove possession of a vendor-signed key without disclosing the key, and provides some unlinkability between attestations. The DAA scheme requires a vendor-issued Endorsement Key as input; the vendor's signature serves as the trust root; and cross-issuer correlation remains possible thru the vendor's view of issued endorsements.

[0016] A further architectural axis distinguishes the prior art systems from the present invention. Identity-creation events in prior art chips occur either at the chip's manufacturing facility (where the manufacturer's controlled environment permits binning out mid-burn failures and presents the chip to the end user only in a

fully-committed state) or by derivation from manufacturing-process variation as in PUF embodiments (where no burn event occurs and the identity is born continuously from first power-on). Owner-initiated burn-time identity creation, in which the chip's owner triggers the identity-creation event in the field on a bench that may be under adversarial power-supply control, faces an atomic-commit problem that the prior art does not address: the chip must present only valid named states at every instant during and after the identity-creation event, including under arbitrary external power interruption, because no controlled facility is available to bin out mid-burn failures and no operator-side recovery procedure may be relied upon. Prior art systems that achieve atomicity by externalizing the identity-creation event to a controlled facility re-introduce the vendor trust path the present invention rejects; prior art systems that achieve atomicity by deriving identity from manufacturing variation require either vendor-controlled enrolment or single-fingerprint operation, neither of which is compatible with the vendor-free domain-separated identity property the present invention requires.

[0017] All identified prior art systems share two foundational architectural assumptions: first, that a party other than the chip's owner participates in the trust path during identity creation; and second, that the device answers identity queries with a structurally correlatable fingerprint, regardless of whether partial unlinkability schemes operate above the wire layer. The present invention rejects both assumptions at the silicon level.

4 BRIEF SUMMARY OF THE INVENTION

[0018] The present invention provides a chip and protocol for vendor-free domain-separated cryptographic device attestation. The chip's permanent identity credential is generated from physical entropy sampled by noise-capture circuitry resident on the chip's silicon at the moment of an owner-initiated one-time provisioning ceremony, denoted herein as the *whakaira*. The unrepeatability of the entropy derives jointly from the owner's choice of triggering moment, unpredictable to any party other than the owner, and from the transient physical state present at and within the chip's silicon at

that moment, unpredictable to every party including the owner; the owner thus controls when the ceremony occurs but not what entropy results, and no party, the owner included, can predict or reproduce the resulting identifier. The sampled phenomena include thermal noise, shot noise, quantum mechanical fluctuations, and ambient electromagnetic and temperature conditions. The captured entropy is conditioned thru an on-chip cryptographic conditioner into the chip's permanent identifier and burned into write-once storage internal to the chip. The permanent identifier itself is never transmitted on any external interface of the chip, in plaintext or in any reversible encoding; subsequent external attestations are cryptographic commitments derived from the permanent identifier under a domain identifier supplied by the requesting application, the commitments being preimage-resistant against recovery of the permanent identifier. No external party witnesses the ceremony, signs the resulting identity, or holds a record of it. The chip's permanent identity therefore has no upstream issuing authority and exists nowhere except inside the chip in which it was created.

[0019] The chip's external interface comprises exactly one bidirectional communication port. In an electrical embodiment, the port comprises a single bidirectional input/output pin coupled to a single conductor; in an optical embodiment, the port comprises a paired light emitter and light detector coupled to a single optical fiber; in a magnetic or inductive embodiment, the port comprises one or more coupling coils; and in capacitive, radio-frequency near-field, and acoustic embodiments, analogous bidirectional coupling structures apply. The protocol carried on the port uses a mirror-XOR line code wherein both endpoints encode each transmitted bit as the exclusive-or of an own-idle bias, a transmit-data value, and the most recently decoded bit received from the other endpoint. The line code produces automatic direct-current balance, free clock recovery, and asymmetric drive that enables tristate idling without bus contention. A leading-edge framing convention announces every message with a single binary one bit transmitted as the channel's transition out of the idle state into active transmission, said leading bit serving as the frame edge for receiver synchronisation. The N-bit message payload follows immediately. Trailing binary zero bits of the payload are not actively transmitted;

the channel returns to its idle state by passive biasing once the last actively driven binary one bit has been transmitted, and the receiver reconstructs trailing zeros by slot-counting against the known message length and decoding non-actively-driven slots as binary zero. The framing cost is therefore exactly one channel slot per message irrespective of payload length, and the framing savings are the trailing zeros of the payload that the sender omits. The channel elapses the payload length plus one slot of time per message in all cases; the framing never introduces a second dedicated framing slot. The decode operation is a slot-counted shift register requiring no framing-symbol detection logic. The per-message channel format carries the payload as masked fields, a masked-message field and a masked-commitment field, each formed by exclusive-or of a value with a session pad derived from a session-shared secret; the masked fields simultaneously provide confidentiality, decryption authority, message authentication, advancement of the session ratchet for forward secrecy, and automatic direct-current balance: five protocol functions delivered by the masked fields rather than by separate dedicated fields, the construction relying only on symmetric and keyed-hash primitives and therefore not on the hardness of any discrete-logarithm or integer-factorisation problem.

[0020] External attestations are produced as keyed cryptographic commitments over the chip's permanent identifier and a domain identifier supplied by the querying application. Two queries with distinct domain identifiers produce cryptographically independent attestations whose correlation requires an information-theoretically infeasible preimage attack on the commitment function. The chip's permanent identifier is not exposed in any attestation. Cross-application correlation of the device is therefore not structurally enabled by the architecture; correlation requires the chip's voluntary participation in producing attestations for both correlating domains, which in turn requires user-mediated consent.

[0021] The chip maintains two distinct identifiers: a permanent identifier denoted *ira* (burned once during the provisioning ceremony) and a per-session secret denoted *wairua* (established once per power-on cycle, at the first handshake following reset, by

the chip's noise-capture circuitry sampling transient physical phenomena present at and within the chip's silicon at the moment of the handshake, and irrecoverably lost upon any interruption of the chip's power supply). Neither identifier alone authorizes operations requiring proof of identity-plus-liveness. A handshake bit from the host doubles as a true-random-number-generation request; the resulting fresh entropy constitutes the chip's response. No separate "request entropy" command exists in the protocol.

[0022] The chip's permanent identifier is stored as multiple cryptographic-inverse pairs in write-once cells, with byzantine quorum required at each chip reset. The provisioning ceremony is atomic with respect to interruption of the chip's external power supply: the chip contains an energy buffer (in some embodiments, an on-die or on-package capacitance; in other embodiments, an external capacitor coupled to the chip's supply pin) and a sensor measuring stored energy against a threshold sufficient to drive the entire identifier-burn sequence to completion at the required burn voltage with margin, and the chip's finite-state machine refuses to begin the burn until the sensor reports that threshold met. Once the burn begins, the energy buffer is isolated from the external supply and supplies the burn to completion regardless of any subsequent external interruption. The entropy-gathering and cryptographic-conditioning sub-steps preceding the burn operate on volatile registers only and write no storage cell; interruption during those sub-steps drops the volatile state and the chip presents the unprovisioned KORE state at next reset. The ceremony therefore presents exactly two protocol-visible states at every instant: KORE before burn-commit, and ORA after burn-completion; no intermediate state is reachable by external power interruption at any time. Single-event upset, manufacturing defect, electromagnetic disturbance, and other post-provisioning corruption are absorbed silently below the byzantine quorum-failure boundary. Below quorum, including the cryptographic-agreement failure in which the valid storage pairs do not agree on a single recovered *ira*, or upon a true-random-number-generation health failure, or upon a non-operational condition of the chip's finite-state machine, the chip enters the NGARO state, wherein wire output is gated off and the wire's idle behaviour is indistinguishable from a chip not physically present. A cryptographic verification failure

on a received protocol message is not an NGARO condition; it is handled by dropping the exchange and re-attempting, as described in the failure-mode section. No error code, failure indicator, or distinguishing signal is emitted on the wire under any failure condition.

[0023] The chip's only clock source is a free-running ring oscillator internal to the chip. The chip has no external clock input, no host-authenticated destruct command, no firmware update path, no method of in-circuit identity replacement, and no in-system mechanism by which an attacker possessing the chip can extract its permanent identity, force it to use entropy from any source other than the chip's noise-capture circuitry sampling at the moment of the triggering event, or substitute a different identity. State transitions are forward-only and enforced by silicon physics rather than by software.

[0024] The resulting device-identity primitive is application-agnostic. Downstream use cases include, but are not limited to: Internet-of-Things device identity without manufacturer involvement after initial provisioning; hardware wallets without factory-injected keys; user-sovereign identity binding as described in the related TOKEN application; automotive secure boot identity surviving ownership transfer; medical device patient-private binding; industrial control vendor-independent attestation; and defense supply-chain post-fabrication substitution detection.

5 BRIEF DESCRIPTION OF THE DRAWINGS

[0025] FIG. 1 is a schematic of a chip configured according to the present invention, captured at the moment of the provisioning ceremony, with every crossing of the chip's package boundary accounted for. A single bidirectional pin, depicted as one bidirectional arrow, carries the owner's triggering request inward and domain-separated commitments outward; a supply rail is interrupted by an isolation switch depicted in its open position, the burn proceeding on energy stored in an on-chip energy buffer; and ambient physical phenomena (electromagnetic, thermal, mechanical) are depicted coupling thru the package as wavy arrows, illustrating that the physics couples thru

the wall while data cannot. Inside the boundary, physical entropy flows from the noise-capture block, thru a cascaded conditioner block (a chaotic-dynamical amplifier followed by a keyed cryptographic hash, labelled HORO and BLAKE3), into an isolated identity store. The identity store is depicted as a double-bordered enclosure holding the chip's two identity primitives as separate sub-stores: the permanent identifier *ira* in write-once storage shown mid-burn, and the per-session secret *wairua* in volatile storage shown with a dashed border to denote loss on power-off, each comprising four inverted pairs. The only path from the identity store to the external pin passes thru a one-way keyed-commitment block, labelled to emit derivatives only; no plaintext path is present in the depicted topology. The owner's triggering event is depicted as an external well-framed provisioning request entering the chip at a moment of the owner's choosing; consistent with the detailed description, the whakaira is triggered only by a complete, well-framed request and not by any stray edge or transient on the pin, and no other external party is depicted. The figure illustrates that the unrepeatability of the entropy derives jointly from the owner's choice of triggering moment and the transient physical state present at and within the chip's silicon at that moment.

[0026] FIG. 2 is a state-machine diagram presented in two panels: an upper panel depicting the provisioning lifecycle and a lower panel depicting the operational degradation ladder, together showing the chip's complete state set and the forward-only transitions among the states. The upper panel depicts the provisioning lifecycle horizontally: the persistent unprovisioned state KORE (write-once storage cells all zero); an owner-press trigger initiating the *whakaira*; a dashed enclosure grouping the transient sub-states KOHI (entropy gathering), HORO (chaotic-dynamical amplification), BLAKE3 (keyed cryptographic hash), and TATARI (randomized pre-burn delay), all operating on volatile registers only; an atomic commit wall depicted as a double vertical stroke; the energy-gated TAHU burn sub-state on the committed side of the wall, entry into TAHU being the irreversible crossing; and the persistent fully-provisioned state ORA. A dashed revert path from the transient enclosure back to KORE depicts that interruption during any transient sub-state presents KORE at next reset, such that exactly two states persist

across a reset during provisioning, KORE before commit and ORA after. The lower panel depicts the operational degradation ladder vertically: ORA (four of four pairs valid) descending thru HARA (three of four pairs valid) and WHARA (two of four pairs valid, at byzantine minimum) by single-pair-loss transitions, and thence to the silent failure state NGARO upon quorum failure; a side annotation shows that NGARO is also entered upon *ira* cryptographic-agreement failure, true-random-number-generation health failure, or any non-operational condition of the chip's finite-state machine, these being chip-integrity failures, whereas a verification failure on a received message is handled by drop-and-retry and does not enter NGARO. The diagram annotates the two-bit chip-state encoding (KORE = binary 00, WHARA = binary 01, HARA = binary 10, ORA = binary 11) and indicates that NGARO is not assigned a numeric code because the chip in NGARO state does not transmit on the wire, the NGARO state being recoverable only by power-on reset and only if the underlying failure condition is no longer present.

[0027] FIG. 3 is a two-panel circuit schematic showing the encode and decode logic of the mirror-XOR line code at the chip endpoint and at the host endpoint, each endpoint comprising a three-input exclusive-or stage (equivalently realisable as two cascaded two-input exclusive-or gates) fed by an own-idle bias signal, a transmit-data signal, and a most-recently-decoded-bit-from-the-other-endpoint signal, together with a slot counter for clock recovery and a tristate-enable signal driving the channel port. The figure further depicts a respective pull-down element at each endpoint and the absence of any drive-low capability at either endpoint. Below the schematic, a worked timing example over a short slot sequence depicts three lanes (enclave drive, host drive, and the resulting composite wire) and isolates the two minimal signalling events, highlighting that because the endpoints hold opposite idle biases the same data bit is signalled by opposite physical gestures: a one from the host (idle low) is an active drive that breaks the line upward, while a one from the enclave (idle high) is a release that lets the line fall to its pull-down, a silence that breaks the line downward.

[0028] FIG. 4 is a full 80-slot full-duplex timing diagram of one complete exchange carrying an ASCII payload from the host endpoint while the enclave endpoint transmits

concurrently, depicting three lanes: the host transmit stream, the enclave transmit stream, and the resulting single conductor that is their mirror-XOR superposition. The figure illustrates that each endpoint's transmit stream is individually unbalanced between high and low logic levels (15 high of 80 for the host, 25 high of 80 for the enclave in the depicted exchange), while the composite conductor is exactly direct-current balanced (40 high, 40 low), the balance emerging from the line code rather than from any balance encoding, and that the enclave's idle bias being the complement of the host's renders the two message representations bitwise inverses.

[0029] FIG. 5 is a per-message wire-format diagram showing the sequential field composition of one authenticated message of the protocol as a single horizontal bar: a single leading binary one bit constituting the channel's transition out of the idle state; a masked-message field X_1 immediately following the leading bit; a masked-commitment field X_2 occupying the bit positions thereafter; and a dashed trailing region, indicated by a separate arrow, in which any binary zero bits at the end of the payload are not actively transmitted and are reconstructed by the receiver by slot-counting against the known message length. The two fields are drawn equal in width for clarity; in general each field may be of any length independently, the widths shown being illustrative and not limiting. The proportions of the bar illustrate that the leading frame bit occupies but a single channel slot while the payload fields occupy the remainder, the framing introducing no dedicated end-of-message marker, length prefix, or inter-message gap. The five protocol functions simultaneously performed by the masked fields, namely confidentiality, decryption authority, message authentication, advancement of the session ratchet for forward secrecy, and automatic direct-current balance, are described in the detailed description in connection with these fields rather than annotated on the bar.

[0030] FIG. 6 is a diagram of the chip's write-once storage array, illustrating the inverted-pair layout of four storage pairs, each pair comprising one cell storing the permanent identifier and one cell storing the bitwise complement of the permanent identifier. The figure further depicts the per-pair validity check comprising a bitwise exclusive-or of the two cells and an all-ones comparator, the byzantine classifier

comprising a count of valid pairs and a minimum-quorum comparator, and a cryptographic-agreement check that all valid pairs recover the same permanent identifier value. A fault-injection annotation illustrates a representative failure mode in which a single bit flip in one cell of one pair invalidates that pair while the remaining three pairs satisfy byzantine quorum, such that the chip continues to operate in the HARA state with the permanent identifier intact.

[0031] FIG. 7 is a flow diagram of the dispatch by which the chip resolves a received message, in particular how it determines, without any plaintext indicator on the channel, whether the message is a chain-reset request or one of an open-ended set of authenticated requests, and for the latter which request it is. A message arrives as a leading frame bit followed by a payload. A single short-circuit is taken first: if the payload is all zero, the message is the reserved chain-reset request, herein the *blip*, and the chip resets the chain (samples fresh entropy, re-initialises the chain state from that entropy, and emits the chain-initialization response). Otherwise the chip performs a single equality check, evaluated for all options in parallel. Each option is an expected value of the form $X_2 = c \oplus H_{resp}$, where H_{resp} is a pad derived from the current chain state and c is the option's candidate value; the received field X_2 is compared against every option's expected value simultaneously, and exactly one option matches or none does, a non-match causing the message to be dropped. The figure depicts representative options, each shown with its candidate: a *wairua* request ($c = K$, where $K = BLAKE3(m_{req})$ and $m_{req} = X_1 \oplus H_{req}$), an *ira* request ($c = \overline{K}$, the bitwise complement, so that the *wairua* and *ira* candidates differ in every bit and at most one can match), a health request ($c = BLAKE3$ of the te reo Māori phrase "Kei te ora koe?", whose response is the chip's packed state vector), and a dashed open-ended option denoting any further selector, each the BLAKE3 of a plaintext phrase chosen by the application; the selector enumeration is unbounded and no plaintext selector appears on the wire. The figure further illustrates that, because the chain is loaded with a random value at power-on that is not disclosed to the host, no received message matches any option until the host has issued a blip to reset the chain to a known state; an unreset

chain therefore yields only non-matches and drops, so no separate “chain established” test is required, and the absence of such a test is reflected in the diagram having no corresponding branch.

6 DETAILED DESCRIPTION OF THE INVENTION

6.1 Terminology

[0032] The present specification uses certain te reo Māori terms for the chip’s identity primitives, ceremony events, and operational states. A typographic convention distinguishes the two categories of term: names set in capital letters (KOHI, HORO, BLAKE3, TATARI, TAHU, KORE, ORA, HARA, WHARA, NGARO, and the message designator Q_HEALTH) denote finite-state-machine states and protocol constants of the chip’s implementation, following the hardware-description-language convention of capitalised state identifiers; names set in lowercase italics (*whakaira*, *ira*, *wairua*, *motuhake*) denote the language-level primitives themselves (the ceremony, the identifiers, and the test domain), which exist independently of any particular state machine that implements them. The terms are used with the following meanings: *whakaira* denotes the ceremony in which the chip’s permanent identifier is brought into existence from physical entropy sampled by noise-capture circuitry resident on the chip’s silicon at the moment of an owner-initiated triggering event; the *whakaira* is owner-initiated, irreversible, and witnessed only by the chip itself. *ira* denotes the permanent identifier resulting from the *whakaira*, and serves as the chip’s lifelong device identity. *wairua* denotes the per-session secret drawn from fresh physical entropy at the first handshake following power-on reset, persisting unchanged in volatile registers until power interruption, and is distinct from *ira*. KOHI denotes a sub-step of the *whakaira* comprising gathering entropy from on-chip true-random-number-generation primitives. HORO denotes a sub-step of the *whakaira* comprising chaotic-dynamical amplification of the gathered entropy thru a fixed-point iterated map with data-dependent operation selection, the te reo Māori word signifying swift motion, a landslide, or to swallow

whole, and reflecting the rapid information-destroying mixing performed by the stage. TATARI denotes a sub-step of the *whakaira* comprising a randomized pre-burn delay whose duration is derived deterministically from the conditioned identifier candidate, the te reo Māori word signifying waiting or delay; the sub-step is interposed between cryptographic conditioning and write-once burn to render the burn-onset moment unpredictable, defeating attacks that must be time-aligned to the burn instant, such as a timed fault-injection disturbance. TAHU denotes a sub-step of the *whakaira* comprising writing the conditioned entropy to write-once storage. KORE denotes the chip's pre-provisioning operational state; the te reo Māori word signifies the primordial void or absence of form. ORA denotes the chip's fully operational state; the te reo Māori word signifies the condition of being alive and well, and corresponds to the chip with all four redundancy pairs verified. HARA denotes a single-fault-tolerated operational state; the te reo Māori word signifies a breach, fault, or error, and corresponds to the chip with three redundancy pairs valid and one pair invalid. WHARA denotes a two-fault-tolerated operational state at minimum byzantine quorum; the te reo Māori word signifies a bruise or injury, and corresponds to the chip with two redundancy pairs valid and two pairs invalid. NGARO denotes the silent failure state; the te reo Māori word signifies the condition of being lost, vanished, or silenced. *motuhake* denotes a canonical test domain identifier used in protocol exercises.

[0033] The chip's five operational state codes are designated as follows. KORE designates the unprovisioned state, in which no *whakaira* has been performed and the chip has no identity, recognised by a byzantine quorum of blank storage groups (groups whose cells are both all-zero) rather than by requiring the entire storage array to read all-zero, so that a stuck or flipped bit on an as-yet-unprovisioned chip does not prevent the chip from being recognised as unprovisioned. ORA designates the state in which the *whakaira* has completed and the chip's identity is recoverable from all four redundancy pairs. HARA designates the state in which one redundancy pair has become invalid and the chip's identity is recoverable from the remaining three valid pairs. WHARA designates the state in which two redundancy pairs have become invalid and

the chip's identity is recoverable at minimum byzantine quorum from the remaining two valid pairs. NGARO designates the state entered upon any of the following chip-integrity failures: byzantine quorum failure of the permanent-identifier storage (including a cryptographic-agreement failure in which the valid storage pairs do not agree on a single recovered *ira*), true-random-number-generation health failure, or any non-operational condition of the chip's finite-state machine. A verification failure on a received protocol message is not an NGARO condition and is handled instead by dropping the exchange and re-attempting. In the NGARO state, the wire output is gated off and the wire's behaviour is indistinguishable from that of a chip not physically present.

[0034] The four wire-reportable state codes are encoded as a two-bit field with the numeric value equal to the chip's vitality on the four-pair scale, with the value increasing as the chip's identity-attesting capability increases. The encoding is: KORE = binary 00, WHARA = binary 01, HARA = binary 10, ORA = binary 11. The NGARO state is not assigned a numeric code because the chip in NGARO state does not transmit on the wire; the receiving host infers NGARO from the absence of any response within the protocol's timeout window, the absence being indistinguishable at the wire from a chip not physically present.

6.2 The Provisioning Ceremony

[0035] The chip is shipped from the manufacturer to the owner in the KORE state. The chip's write-once storage cells are nominally all zero, and the chip recognises the KORE state by a byzantine quorum of blank storage groups, so that a fabrication defect leaving a stuck bit in one group does not prevent recognition. The chip's finite-state machine reports a chip_state value of KORE. The chip has no permanent identity in this state. The chip cannot produce an identity-bearing attestation in this state because no *ira* has been written; the chip remains able to draw *wairua* from its noise-capture circuitry and to emit fresh entropy in response to a handshake, those operations being independent of *ira*. The chip's manufacturer, having shipped the chip in this state, possesses no information concerning the chip's identity, because the chip's identity does not exist.

[0036] An owner of the chip initiates the *whakaira* by transmitting a structured attestation-request message to the chip's external bidirectional pin under the protocol's framing convention, namely a leading-edge frame bit followed by the message payload. The chip's finite-state machine accepts the request only after the receiver has decoded a complete, well-framed protocol message; transient electrical disturbances on the pin, including electrostatic discharge events, capacitive coupling artefacts, and isolated edges arising from connector insertion or wire handling, do not satisfy the framing convention and are rejected by the receiver without state transition. The *whakaira* trigger is specifically the first permanent-mode attestation request received while the chip is in the KORE state: a well-framed request whose cryptographic invariant, described in connection with the identity dichotomy below, selects the permanent identifier as the active commitment key. Session-mode requests, handshakes, and state queries received while the chip is in the KORE state are served from *wairua* in the ordinary way and do not trigger the *whakaira*. When the chip is in any provisioned state, a permanent-mode attestation request is served as an ordinary attestation without re-entering the *whakaira*. The energy-buffer threshold gate described separately herein operates as an independent second condition: the finite-state machine refuses to begin the TAHU burn until the on-chip energy sensor reports stored energy above the burn-completion threshold, regardless of the framing-validated request having been received.

[0037] In the KOHI state, the chip's noise-capture circuitry samples transient physical phenomena present at and within the chip's silicon at the moment of the sampling. The phenomena include, in various embodiments, thermal noise (Johnson-Nyquist noise arising from the silicon's instantaneous temperature), shot noise (arising from the quantization of electron flow at the moment of sampling), quantum mechanical fluctuations in metastable resolution, ambient electromagnetic field conditions, and other transient environmental physical noise sources present at and within the chip. The unrepeatability of the resulting entropy derives jointly from the owner's choice of triggering moment, which fixes *when* the sampling occurs and is unpredictable to

any party other than the owner, and from the unrepeatable transient physical state present at that moment, which fixes *what* entropy results and is unpredictable to every party, the owner included. The owner therefore controls the timing of the ceremony but not its outcome; no party, the owner among them, can predict or reproduce the resulting identifier. The chip's noise-capture circuitry comprises, in some embodiments, a plurality of free-running ring oscillators sampled thru statistical entropy-extraction tests including, in some embodiments, the repetition count test, the adaptive proportion test, and an iterated derivative test specified by NIST Special Publication 800-90B; in some embodiments, one or more transition-effect ring oscillators; in some embodiments, one or more metastable arbitration circuits; in some embodiments, one or more subthreshold-inverter shot-noise samplers or thermal-noise amplifiers; or in some embodiments, any combination of the foregoing operated in parallel. The combination of capture circuits is a matter of implementation choice; the requirement is that the sampling be performed by noise-capture circuitry resident on the chip's silicon, and that the chip's protocol expose no pathway by which an external party can supply chosen bits as protocol data that the conditioner accepts as entropy. The chip's noise-capture circuitry necessarily transduces physical phenomena originating outside the silicon, including ambient temperature, ambient electromagnetic field conditions, and external mechanical or acoustic disturbances coupling into the silicon; such environmental coupling is a physical mechanism by which entropy arises and is not excluded. Physical noise coupling onto the chip's external pin from the pin's own electromagnetic environment is likewise not excluded and may contribute a typically small amount of additional entropy; what is excluded is not physical coupling of any kind but any pathway by which an external party could deliver bits of *intended* entropy to the chip as protocol data, the chip's protocol providing no command, register, or state in which the on-chip conditioner accepts bits arriving at the pin as a substitute for, or chosen contribution to, the entropy sampled by the on-chip noise-capture circuitry. The security property is therefore not the absence of all external physical influence, which is physically unattainable, but the absence of any protocol-level channel by which an adversary could substitute or bias the entropy with values of the adversary's choosing.

[0038] The chip gathers entropy until on-chip health diagnostics confirm sufficient entropy quality. If the health diagnostics report failure, the chip enters the NGARO state and the *whakaira* is aborted. In this case, the chip silently refuses to provision rather than provisioning with insufficient entropy. The chip cannot be coerced into substituting attacker-supplied bits for its on-chip noise sources, because the chip's protocol provides no pathway by which bits arriving at the external pin reach the on-chip conditioner. If the silicon's true-random-number generator is broken, the chip refuses to claim an identity it cannot defensibly attest under.

[0039] Upon successful completion of KOHI, the finite-state machine transitions to the HORO state. In this state, the entropy gathered in KOHI is fed to a chaotic-dynamical amplifier comprising a fixed-point iterated map over a plurality of state buckets, in some embodiments sixteen buckets each carrying a sixteen-bit fraction and an eight-bit exponent, mixed across a plurality of rounds, in some embodiments sixteen rounds. Each round applies to each bucket one operation selected from a menu of distinct lightweight arithmetic-logic operations by an index that is itself derived from the bucket's current value, said data-dependent operation selection producing a combinatorial path explosion that complements the data-dependent shift amounts and the per-round round-constants. A subset of the operation menu is information-lossy, in some embodiments approximately fourteen of thirty-two operations being lossy, such that the amplifier cumulatively destroys on the order of several hundred to several thousand bits of internal state across its rounds, said destruction occurring within a state space populated by entropy substantially in excess of the destruction bound. The HORO stage thereby preserves a high lower bound on retained min-entropy while ensuring that any partial recovery of the stage's output is computationally precluded from inverting to the stage's input, and produces a chaotically mixed intermediate value as input to the cryptographic-conditioning sub-state that follows. The chaotic amplifier circuit is part of the chip's mask design and operates on volatile registers only.

[0040] Upon successful completion of HORO, the finite-state machine transitions to the BLAKE3 state. In this state, the chaotically mixed intermediate value emerging

from HORO is concatenated with the original entropy gathered in KOHI, and the concatenation is conditioned thru an on-chip keyed cryptographic hash function implemented in hardware logic. In some embodiments, the hash function is BLAKE3 operating in keyed mode. The output of the conditioner is a 256-bit cryptographic-quality value, deterministic with respect to the gathered entropy and resistant to bias from any non-uniformity in the entropy source. The conditioner circuit is part of the chip's mask design and is therefore known to the chip's manufacturer; however, the manufacturer does not know the conditioner's input on any specific chip, because the input is the entropy gathered in KOHI on that specific chip and the chaotic transform performed in HORO on that specific chip.

[0041] Upon completion of BLAKE3 conditioning, the finite-state machine transitions to the TATARI state, holding the conditioned identifier candidate in volatile registers for a randomized delay before the burn. The randomized delay renders the burn-onset moment unpredictable to an external party. This defeats the class of attacks that must be precisely time-aligned to the burn instant to succeed: a fault-injection disturbance applied at the exact moment the storage cells are written, such as a sub-nanosecond high-voltage electromagnetic pulse, a focused optical or charged-particle pulse, or a mechanical or acoustic shock, timed to corrupt the write as it occurs. Such an attack does not act thru the chip's power source and is therefore not addressed by the energy-buffer isolation described below; its sole requirement is knowledge of when the burn fires, and the randomized delay denies that knowledge, forcing the attacker to gamble the disturbance somewhere within a window many orders of magnitude wider than the burn pulse it must hit. Upon expiry of the delay, the finite-state machine evaluates the energy-buffer threshold gate before isolating power and committing to TAHU.

[0042] The energy-buffer threshold gate is the final precondition of the TAHU burn. The chip's mask design incorporates an energy buffer coupled to the chip's supply rail and a sensor measuring the buffer's stored energy. In some embodiments the buffer comprises an on-die or on-package capacitance; in other embodiments the buffer

comprises an external capacitor coupled to the chip's supply pin and dedicated to the burn. The sensor's threshold is selected to exceed the integrated energy required to drive the complete TAHU write sequence (comprising the programming pulse at the required burn voltage applied sequentially or concurrently to every storage cell of the inverted-pair layout) by a design margin sufficient to absorb worst-case process, voltage, and temperature variation in the burn-cell programming current, plus a further margin against measurement uncertainty in the sensor itself. Upon expiry of the TATARI delay, the finite-state machine samples the sensor and refuses to transition to the TAHU state unless the sensor reports stored energy above the threshold. If the sensor reports stored energy below the threshold, the finite-state machine drops the conditioned identifier candidate from its volatile registers and presents the KORE state at next reset, the chip's owner being free to re-attempt the *whakaira* after replenishing the energy buffer; no write-once storage cell is touched and no information about the discarded candidate persists. If the sensor reports stored energy above the threshold, the finite-state machine performs two electrically simultaneous actions: it isolates the energy buffer from the chip's external supply rail thru a switch element in the chip's mask design that prevents subsequent external supply interruption from draining the buffer, and it transitions to the TAHU state. The isolation switch remains engaged until the TAHU sequence completes. The combined effect of the threshold-with-margin check and the supply isolation is that, once TAHU begins, the burn is electrically guaranteed to run to completion against any external power interruption strategy: the buffer alone supplies the burn, the buffer was verified to contain more energy than the burn consumes plus margin, and the buffer is no longer connected to anything an attacker can pull down. This guarantee is specific to the power-supply attack vector; timed fault-injection disturbances that do not act thru the power supply are addressed separately by the randomized burn-onset timing of the TATARI state described above, and any corruption that nonetheless reaches the storage array is absorbed at read time by the byzantine quorum classifier described in a subsequent section. The three mechanisms are independent: the energy buffer guarantees completion against power interruption, the randomized delay denies the timing required by injection attacks, and

the quorum classifier rejects a partially-corrupted array at next reset.

[0043] In the TAHU state, the conditioned 256-bit value is written to the chip's write-once storage cells according to the inverted-pair layout described in connection with FIG. 6. The write operation is electrically irreversible: the cells transition from an unwritten state to a written state, and the chip's mask design provides no path by which they can return to the unwritten state. Isolation during TAHU is limited to the write-once storage section and its burn-current path; the remainder of the chip continues to operate from the energy buffer or external supply and may continue to service *wairua*-based handshake and boot-only attestation requests on the external pin during the burn, said requests being independent of the stored identifier. Device-specific attestation, which requires the stored identifier, becomes available only upon the FSM's subsequent transition to the ORA state. The stored identifier is at no time transmitted on the external pin in plaintext or in any reversible encoding; ORA-state responses to device-specific attestation requests are cryptographic commitments derived from the stored identifier under a domain identifier supplied by the requesting application, said commitments being preimage-resistant against recovery of the stored identifier.

[0044] Upon completion of TAHU, the finite-state machine transitions to the ORA state. The chip is provisioned. The chip's identity exists. The chip will not reveal the identity to any external party, including the chip's owner, the chip's manufacturer, or any authority that may compel disclosure at any future time. The identity is a quantity inside silicon that no party outside the silicon observes.

6.3 The Single Bidirectional Interface

[0045] The chip's external interface comprises exactly one bidirectional communication port. The chip does not include a separate clock input, a debug interface, a JTAG tap, an in-circuit firmware update mechanism, an out-of-band manufacturer access path, a destruct port, or any other interface that crosses the chip's package boundary. The architectural decision to provide a single bidirectional port is not aesthetic; each interface

absent from the chip is a class of attack absent from the chip.

[0046] The physical embodiment of the single bidirectional port may vary by application. In an electrical embodiment, the port comprises a single bidirectional input/output pin coupled to a single conductor, with one or more pull-down resistors (or, in some embodiments, one or more pull-up resistors) coupled to the conductor establishing the conductor's idle electrical state. A single such resistor on the shared conductor is sufficient; in some embodiments, for symmetry of the two endpoints' analog interfaces, a respective pull-down resistor is provided at each endpoint's pin, optionally in series with a respective current-limiting resistor, the parallel combination establishing the same idle state. The number and placement of the biasing resistors is an analog implementation choice and does not alter the line-code semantics, which depend only on the conductor settling to the idle level whenever no endpoint actively drives.

[0047] In some electrical embodiments the chip's pin includes a hardened analog front-end interposed between the external conductor and the chip's core logic, serving both ordinary signal-integrity and electrostatic-discharge purposes and the defensive purpose of attenuating injected fault transients. The front-end comprises, in various embodiments, any one or more of: a series inductive element, such as an inductor or ferrite bead, in line with the pin, opposing the rapid current changes characteristic of an injected glitch or electromagnetic pulse; a shunt capacitive element from the pin to a reference, soaking high-frequency energy and slowing edge rates; and one or more clamp elements, such as a pair of clamp diodes or a transient-voltage-suppression element, bounding the pin excursion to a range between the pin's reference rails. The series inductive and shunt capacitive elements together form a low-pass filter whose passband is selected to accommodate the protocol's slot rate while attenuating sub-nanosecond transients of the kind used in fault-injection attacks, the wide separation between the protocol's modest slot rate and such transient time scales permitting effective attenuation without distorting the signal. In some embodiments the conductor and pin operate in a higher voltage domain than the chip's core logic, for example a nominal five-volt signal swing against a lower-voltage core, the higher swing

providing greater noise margin and greater resistance to line-injected corruption; in such embodiments the pin is coupled to the core logic thru bidirectional level translation, the received signal being level-shifted down to the core logic level on the receive path and the core logic's drive being level-shifted or buffered up to the pin's voltage domain on the transmit path, the level translation being transparent to the line-code semantics. The selection and combination of front-end elements and voltage domains is an analog implementation choice and does not alter the line-code semantics.

[0048] In an optical embodiment, the port comprises a paired light emitter and light detector at the chip side and a corresponding paired emitter and detector at the host side, with both pairs coupled to a single optical fiber operating as a bidirectional channel; the channel's idle state is established by an optical bias source or by passive emitter quiescence with detector thresholding. In a magnetic or inductive embodiment, the port comprises one or more coupling coils, with the channel operating thru near-field magnetic coupling and the channel's idle state established by a quiescent flux level. Capacitive, radio-frequency near-field, and acoustic embodiments employ analogous structures. References herein to "wire," "conductor," "drive," "pull-down," "high logic level," "low logic level," and similar electrical terminology are used to describe the line code clearly in the electrical embodiment; in non-electrical embodiments, the corresponding analog structures and signal states apply, and the term "channel" is used herein when the description applies generally to any physical embodiment.

[0049] The absence of a JTAG interface eliminates in-system invasive probing of the chip's internal state. The absence of a firmware update mechanism eliminates the possibility of future vendor-pushed identity replacement. The absence of a debug interface eliminates any manufacturer access path designated as a service mode. The absence of a destruct pin eliminates any host-authenticated wipe that an attacker possessing the chip could trigger. The absence of an external clock input eliminates clock-glitching attacks against the chip's finite-state machine.

[0050] The chip's only clock source is a free-running ring oscillator internal to the chip. The frequency of the ring oscillator varies with temperature, voltage, and process

variations; the protocol described in the present specification tolerates this variation thru automatic clock recovery in the line code. The host endpoint does not share an external clock signal with the chip. The host endpoint derives the chip's slot timing from observing the chip's wire transitions during the heartbeat phase of the protocol. A single chip can therefore be installed in any host that implements the protocol, regardless of the host's own clock, without provisioning a clock relationship between the chip and the host.

[0051] The chip's finite-state machine transitions are forward-only. State transitions are enforced by combinational logic in the chip's mask design that physically does not provide a backward transition. The transition from KORE to KOHI is unidirectional. The transition from KOHI to HORO is unidirectional. The transition from HORO to BLAKE3 is unidirectional. The transition from BLAKE3 to TATARI is unidirectional. The transition from TATARI to TAHU is unidirectional. The transition from TAHU to ORA is unidirectional. The NGARO state, once entered, is recoverable only by power-on reset and only if the underlying failure condition is no longer present. The chip's mask design provides no software-controlled path by which an attacker could cause the chip to traverse its state graph in reverse, nor any path by which an attacker could leave the chip in a state partially committed between two adjacent FSM states. The pre-burn sub-steps (KOHI, HORO, BLAKE3, TATARI) operate on volatile registers only and present the KORE state at next reset if interrupted; the TAHU sub-step is gated by the energy-buffer threshold and supply-isolation arrangement described above and runs to completion against any external supply manipulation once entered; no other transition writes persistent state. Consequently the chip exposes exactly two persistent post-reset states to any external party: KORE (no *ira* ever committed), and a quorum-determined operational state (ORA, HARA, WHARA, or NGARO) reflecting the committed *ira*. No external action, including power glitching, line glitching, electromagnetic injection, or any combination, can drive the chip into a persistent intermediate or partially-burned state; glitch attacks that attempt to corrupt the evaluation of the chip's logic by perturbing its operating conditions are further addressed by the operating-envelope monitor

described below, which holds the chip inoperative and wire-silent whenever supply voltage, temperature, internal clock frequency, electromagnetic field strength, or incident radiation falls outside the chip’s valid operating range.

6.4 The Mirror-XOR Line Code

[0052] The mirror-XOR line code carries the present protocol over the single bidirectional channel coupled to the chip’s external port. The description below is given for the electrical embodiment; the same logic applies to optical, magnetic, inductive, capacitive, radio-frequency, and acoustic embodiments with corresponding signal states substituted. Time on the channel is divided into uniform slots whose ownership alternates between the two endpoints. Each endpoint maintains a data stream of bits intended for transmission; in idle the data stream is constant zero, and activity is signalled simply by inserting non-zero bits into the stream. In its own slot, an endpoint computes wire_out as the exclusive-or of three values: own_idle, own_data, and other_last_data, where own_idle is a fixed per-endpoint constant (binary 1 at the chip endpoint, binary 0 at the host endpoint), own_data is the next bit in the endpoint’s own data stream, and other_last_data is the bit most recently decoded from the other endpoint. The endpoint actively drives the channel HIGH if wire_out evaluates to 1, and releases the channel to its passive idle level (the pull-down resistor in the electrical embodiment) if wire_out evaluates to 0; the other endpoint never drives during this slot, so the channel state in any slot has a single defined source. In the other endpoint’s slot, an endpoint decodes the observed channel value as the exclusive-or of three values: the observed wire value at the bit’s sample point, the other endpoint’s idle constant other_idle, and own_last_data, the bit the endpoint itself most recently transmitted.

[0053] The asymmetric drive convention has the following consequences. With both data streams at constant zero, the chip endpoint computes wire_out = 1 and drives HIGH in its slots, while the host endpoint computes wire_out = 0 and releases the channel to its pull-down LOW in its slots; the channel idle pattern is therefore the deterministic alternating sequence $1, 0, 1, 0, 1, 0, \ldots$, a self-clocking heartbeat carrying

100% transition density for clock recovery without any active host contribution. The chip endpoint transitioning from HIGH to LOW within one of its own slots (produced by the chip inserting a 1 into its data stream, which under the mirror-XOR rule with stream-idle preconditions evaluates to wire_out = 0 and a release) constitutes the message-start edge. No separate framing symbol is transmitted; a 1 in the data stream is itself the frame edge. The same applies to the host, whose insertion of a 1 produces wire_out = 1 in a host slot, breaking the idle LOW with an active HIGH drive. Both directions therefore operate full-duplex with framing emerging directly from the data, and the XOR compensation against other_last_data restores direct-current balance within at most a two-slot transient following any inserted 1.

[0054] The drive logic at each endpoint computes a drive-enable signal as the logical AND of the endpoint's own slot ownership indicator and the endpoint's computed wire output value. Each endpoint drives the wire to a high logic level when the drive-enable signal is asserted. Each endpoint releases the wire to a high-impedance state when the drive-enable signal is deasserted. The wire is held at a low logic level in the absence of drive by an external pull-down resistor coupled to the conductor. Neither endpoint drives the wire to the low logic level. Bus contention between the endpoints during simultaneous transmission slots is structurally impossible because both endpoints' slot ownership indicators cannot be simultaneously asserted.

[0055] The line code produces three properties without dedicated hardware support. First, direct-current balance over the wire emerges from the encoding. The mirror-XOR compensation term drives the wire toward an equal count of high and low slots: a bit that one endpoint contributes in one polarity is offset by the other endpoint's compensating drive in the following slot, so the running difference between high and low slot counts returns to zero within at most a one-slot transient. Over a window of slots the wire's mean voltage therefore settles at the midpoint between the high and low logic levels, and for statistically uniform payload bits, such as the masked fields of the per-message wire format, the balance over the window is exact in expectation. No bias resistor, decoupling capacitor, or explicit balance encoding is required.

[0056] Second, free clock recovery emerges from the encoding. Each transition in the wire's state is a clock edge for the receiving endpoint. In the heartbeat phase of the protocol, the wire alternates between the high and low logic levels at the slot rate, providing the receiver with a transition per slot for clock synchronisation. During data transmission, the statistical randomness of the data ensures that wire transitions occur with sufficient density to maintain receiver clock lock. The clock recovery hardware comprises one logic gate and a slot counter; no dedicated phase-locked loop, delay-locked loop, or explicit synchronisation pattern is required.

[0057] Third, the asymmetric drive property enables silent-on-corruption behaviour. The chip endpoint never drives the wire to the low logic level. Gating off the chip's drive-enable signal causes the wire to fall to the low logic level under the action of the external pull-down resistor. The chip does not actively transmit a silence signal; the chip merely stops driving. This property enables the NGARO state to produce wire behaviour indistinguishable from the chip's physical absence: the chip stops driving, the wire returns to its pull-down idle, and an external observer cannot distinguish a chip that is present but has stopped driving from a chip that is not present.

[0058] Each endpoint samples the channel state at a sampling rate substantially higher than the slot rate and accumulates the samples across the slot width by means of an up/down counter. The counter is incremented on each sample of the channel at the high logic level and decremented on each sample of the channel at the low logic level. At the slot boundary the receiving endpoint decodes the slot's bit value as the sign of the accumulated counter: a positive accumulated value decodes as binary 1, and a non-positive accumulated value decodes as binary 0. The counter is then reset for the next slot. For binary-valued channel samples, the sign-of-sum operation is mathematically equivalent to a majority vote across the samples of the slot, and is therefore equivalent to a median filter of width equal to the number of samples per slot. The receiver consequently tolerates noise events shorter in duration than one half of the slot width without bit error, irrespective of the noise event's amplitude, because such noise events flip fewer than one half of the slot's samples and therefore cannot change

the sign of the accumulated counter. The sampling architecture comprises one up/down counter and one sign-bit register per endpoint; no analog filter, equalizer, or feedback loop is required. The same architecture is used at both endpoints.

[0059] The per-message wire format carries the payload as two masked fields: a masked-message field X_1 and a masked-commitment field X_2. Letting m denote the payload value of the exchange and H_{req}, H_{resp} denote two session pads derived from the session-shared secret, the masked-message field is $X_1 = m \oplus H_{req}$ and the masked-commitment field is $X_2 = BLAKE3(m) \oplus H_{resp}$. These masked fields simultaneously perform five protocol functions that conventional cryptographic protocols deliver via separate fields. The construction uses only symmetric exclusive-or masking and keyed cryptographic hashing; it does not employ public-key key agreement, and therefore depends on no discrete-logarithm or integer-factorisation assumption and retains its security against a quantum adversary up to the generic square-root bound on the underlying primitives.

[0060] The first function is confidentiality. The payload value m is carried only in masked form $X_1 = m \oplus H_{req}$; an observer of the channel without the session pad H_{req} obtains no information about m, the masking being a one-time-pad exclusive-or under a pad freshly derived from the session-shared secret.

[0061] The second function is decryption authority. A receiver holding the session pad recovers the payload value as $m = X_1 \oplus H_{req}$. Possession of the correct H_{req}, derived from the session chain, is precisely what authorizes recovery; no separate key-transport field is transmitted.

[0062] The third function is message authentication. The receiver recomputes $BLAKE3(m)$ over the recovered payload value and verifies it against $X_2 \oplus H_{resp}$. Any modification of either masked field, or any divergence of the receiver's session pads from the sender's, causes the verification to fail and the message to be dropped, so the masked-commitment field authenticates the masked-message field and binds it to the session.

[0063] The fourth function is advancement of the session ratchet, providing forward secrecy. The session pads for the next exchange are derived from $BLAKE3(m)$ of the present exchange, so each exchange consumes fresh pads and no pad is reused; compromise of the session-shared secret at one point does not retroactively expose payloads of prior exchanges, whose pads were derived from payload values already mixed forward and not recoverable from the current state.

[0064] A single binary one bit is transmitted before every message as the wire's transition out of the idle state into active transmission, the receiver synchronising message-start detection to said transition. This leading frame bit is provided by the framing convention of claim 4 and is not one of the functions of the masked fields; it is not a bit of the message payload, occupying the slot immediately preceding the first bit of the masked-message field.

[0065] Concomitantly, trailing zeros of the message payload are not actively driven onto the channel. The sender ceases active transmission after the last actively driven binary one bit of the payload, and the channel returns to its idle low state by passive biasing. The receiver reconstructs the trailing zeros by slot-counting N slots after the leading frame bit, where N is the known fixed payload length, and decoding each non-actively-driven slot as binary zero. The decode operation is therefore a slot-counted shift register without dedicated framing-symbol detection logic: each slot is sampled, and any slot not actively driven by the remote endpoint is decoded as zero.

[0066] The combination of the leading frame bit with the omitted-trailing-zeros convention bounds the framing overhead at one channel slot per message, and the channel elapses the payload length plus one slot of time in all cases. The framing never introduces a second dedicated framing slot beyond the leading frame bit; in particular, no trailing end-of-message marker, no length prefix, and no inter-message gap signaling is present in the wire-level protocol. The same channel-idle state at the end of transmission also serves as the silent-on-corruption indicator described in the failure-mode subsection: a chip in the NGARO state and a chip having completed transmission both present an undriven channel; the receiver disambiguates by

message-length accounting and timeout, not by additional wire signaling.

[0067] The fifth function is automatic direct-current balance. Each masked field is the exclusive-or of a value with a cryptographic-quality pad and therefore has an approximately uniform bit distribution. The 256 bits of each of X_1 and X_2 contribute approximately 128 high and 128 low values to the line code, supplying the statistical uniformity on which the mirror-XOR direct-current balance relies, without a dedicated balance encoding field.

[0068] No prior art cryptographic protocol combines all five of the foregoing functions in the masked payload fields at the wire level. Individual protocols provide subsets of the five functions thru separate dedicated fields. The combination of all five functions in the masked fields, using only symmetric and keyed-hash primitives, is novel.

6.5 Domain-Separated Per-Application Attestation

[0069] The chip produces an attestation in response to a query received from a requesting application. The query includes a domain identifier supplied by the requesting application. The domain identifier may be any value the application selects, such as a service uniform resource locator, an application name, a randomly generated handle, a hash of the application's identity, or a domain-specific identifier such as a use-case label. The chip does not constrain the structure of the domain identifier.

[0070] The attestation produced by the chip is a keyed cryptographic commitment derived under an active commitment key and computed over an input that includes the domain identifier supplied by the requester. The active commitment key is selected from among *ira* and *wairua* according to the identity dichotomy described in a subsequent section. The active commitment key is not consumed by a single cryptographic hash function alone, but by the same cascaded HORO-plus-BLAKE3 conditioner described in connection with the provisioning ceremony: the active key, concatenated with the supplied domain identifier and any session-specific freshness inputs, is first passed thru the chaotic-dynamical amplifier (HORO) for chaotic mixing with information-loss,

and the amplifier's output is then concatenated with the original input and conditioned thru a keyed cryptographic hash function (in some embodiments BLAKE3 operating in keyed mode) to produce the 256-bit attestation. The cascaded conditioner is used in lieu of a single hash construction so that the security of the attestation against recovery of the active commitment key does not rest on the preimage-resistance of any one cryptographic primitive in isolation; an adversary attempting to recover the active key from a published attestation must defeat both the chaotic-dynamical amplifier's path-explosion plus information-loss combination and the cryptographic hash's preimage-resistance, the two being independent security arguments.

[0071] The requesting application verifies the attestation by recomputing the keyed cryptographic commitment under a previously-established public commitment derived from the chip. The chip's public commitment is a 256-bit value derived from the chip's permanent identifier and shared at the time the chip was first registered with the application. The public commitment does not reveal the chip's permanent identifier; it is sufficient for the application to verify subsequent attestations produced by the chip.

[0072] Two queries with distinct domain identifiers produce cryptographically independent attestations. To correlate the chip across two such queries, a third party would be required to compute one query's attestation from the other's, which requires either a preimage attack on the keyed cryptographic commitment (which is information-theoretically infeasible) or the chip's voluntary participation in producing both attestations. The chip produces an attestation only in response to a query that supplies the corresponding domain identifier. The chip does not produce attestations spontaneously. The chip's user controls which queries are delivered to the chip. Cross-application correlation of the chip is therefore not a structural property of the architecture; correlation requires the user to consent to allowing both correlating applications to query the chip.

[0073] This property distinguishes the present invention from prior art device-identity primitives commercially available in 2026. Trusted Platform Module endorsement keys, Apple Secure Enclave unique identifiers, Google Titan keys, and per-chip identifiers

from hardware identity element manufacturers including STMicroelectronics, NXP, Infineon, and Microchip, answer device-identity queries with values that any third party can correlate by inspection. The present invention answers device-identity queries with values that are cryptographically independent across all domain identifiers. The privacy property is enforced at the silicon level rather than by a policy layer above the silicon.

[0074] The Q_HEALTH state-query exchange is a specific application of the in-session messaging mechanism, distinct from the attestation mechanism described above. Q_HEALTH is a request for the chip's current operational state vector. It is not an attestation, does not invoke *ira*, does not reveal the chip's permanent identity, and produces a response of the same wire format and the same size as any other in-session message. The Q_HEALTH response carries the chip's packed operational state vector (including the two-bit chip_state field and any further per-redundancy-pair validity bits) within the response payload, the payload being authenticated and bound to the active session by the same chain-state mechanism that authenticates every other response in the session. A verifying application correctly receives the state vector only by decoding the response against its expected session chain state; if the application's chain state has diverged from the chip's, or if an attacker has interposed an impersonating endpoint, the decode yields uniformly distributed garbage indistinguishable from a corrupted message and the application takes its ordinary session-corruption recovery path. The chip's identity is bound to the session at the time of session establishment, by the attestation mechanism described in the preceding subsection; Q_HEALTH inherits that binding without re-establishing it per query, which is what permits the response to remain identical in size to any other in-session message rather than carrying a per-response identity commitment. The Q_HEALTH embodiment illustrates that in-session state reporting and per-application attestation are independent mechanisms in the present protocol: an application that has already established a session with a given chip does not need to repeat the attestation step to obtain trustworthy state, and conversely a Q_HEALTH response carries no information usable by any party that has not already established the session.

6.6 The Permanent-Plus-Session Identity Dichotomy

[0075] The chip maintains two distinct identifier values inside the chip. The first identifier, *ira*, is generated by the provisioning ceremony described above and is stored in the chip's write-once storage. The second identifier, *wairua*, resides in the chip's volatile registers. *Wairua* is established once per power-on cycle, at the chip's first handshake following reset, by the chip's noise-capture circuitry sampling transient physical phenomena present at and within the chip's silicon at the moment of the handshake. Once established, *wairua* persists unchanged for the full power-on lifetime of the chip; subsequent handshakes within the same power-on cycle deliver fresh entropy to the host and refresh per-message session chain state, but do not replace *wairua*. The chip's volatile registers, including those holding *wairua*, lose their contents irrecoverably upon any interruption of the chip's external power supply, the chip's mask design providing no battery, no on-die retention element, and no mechanism by which *wairua* can persist across a power-on reset.

[0076] *Wairua* is held in the volatile registers using the same inverted-pair discipline as the permanent identifier: a plurality of register groups, in some embodiments four, each holding *wairua* in one register and the bitwise complement in the paired register, verified at use by the same exclusive-or validity check applied to the write-once storage groups. The mirrored layout converts single-event upsets of the volatile registers into detected validity failures rather than silent corruption. Upon a validity failure of the *wairua* groups during a power-on cycle, the chip ceases responding on its external interface for the remainder of the power-on cycle; recovery requires interruption of the chip's power supply followed by a fresh handshake, both endpoints re-deriving session state in lock-step. The chip performs no silent re-derivation of a corrupted *wairua*, because session state derived from a silently re-derived secret would diverge undetectably from the host endpoint's session state; the forced power cycle converts the corruption into an observable session break.

[0077] The two identifiers carry complementary properties, each insufficient on its own

and each providing a distinct security guarantee in combination. *ira* is permanent and identical across every session of the chip's life; *wairua* is fresh per session and stable within a session, and is cryptographically independent of any other session's *wairua*. The properties that follow from each identifier's behaviour are therefore distinct.

[0078] *ira* alone, used as the sole key for attestations, is replay-vulnerable: any attestation the chip produces under *ira* alone may be captured and presented again to any application that holds the chip's public commitment, with no mechanism distinguishing a fresh attestation from a stored one. Without per-session freshness, attestations under *ira* alone are durable artifacts subject to lift-and-reuse attacks.

[0079] *wairua* alone, used as the sole key for attestations, provides freshness but not cross-session identity. Across two distinct sessions, the *wairua* values are independent samples drawn from the chip's noise-capture circuitry and conditioned to cryptographic-quality output, and are by construction unlinked to each other and to any persistent quantity. A verifying application receiving a *wairua*-keyed attestation in session A and a *wairua*-keyed attestation in session B can derive no relation between the two from their attestation values, and therefore cannot conclude that the producer of attestation A is the same chip as the producer of attestation B; any true-random-number generator coupled with the same conditioner could have produced statistically indistinguishable values.

[0080] Within a single session, however, *wairua* is stable, and this stability is itself a security-relevant primitive of independent utility. A verifying application that issues the same query to the chip at two distinct points in time during the same session receives the same response, the response being a deterministic function of *wairua* (constant within the session) and the query (identical between the two issuances). A subsequent identical query that returns the identical response is therefore cryptographic proof that the chip's volatile state has been continuously preserved between the two issuances: no interruption of the chip's external power supply has discharged the chip's volatile registers, and no replacement of the chip with a different physical chip has occurred. The same property holds in reverse: an identical query that returns

a different response is cryptographic proof that the chip's session has been broken, whether by power interruption or by physical replacement. This session-continuity proof property is a per-session liveness primitive useful for, among other applications, satellite or other remote-deployed device integrity audit (in which a ground station wishes to confirm a deployed device has not been reset, replaced, or maliciously rebooted since deployment), and tamper-evident continuous operation of high-assurance systems (in which any interruption of the chip's session evidences the tampering). The converse use of the same *wairua* stability, in which a capability is bound *to* a session's *wairua* so that loss of the session destroys the capability and the loss is itself provable, is the hardware-rooted provable-erasure primitive described in a subsequent subsection; the content-rental example, in which content is encrypted under the session *wairua* and a return is proven by the chip thereafter exhibiting an unchanged *ira* but a different *wairua*, belongs to that converse use rather than to the continuity proof described here.

[0081] A permanent-mode attestation combines the two identifiers in a single derivation: the commitment is computed thru the cascaded HORO-plus-BLAKE3 conditioner over *ira*, per-session freshness derived from *wairua*, and the domain identifier supplied by the requester. The combination provides the security properties of each constituent: *ira*'s permanent-identity binding ensures that attestations across sessions authenticate to a single chip's public commitment, and the *wairua*-derived freshness ensures both that each session's attestations are unique to that session, defeating cross-session replay, and that intra-session repetition of a query yields the session-continuity proof described above. A session-mode attestation is keyed from *wairua* and the session chain state alone and carries no permanent-identity binding.

[0082] The active commitment key for each specific exchange is selected by a cryptographic invariant carried in the host's message rather than by a plaintext mode indicator. In some embodiments, the cryptographic invariant comprises a comparison of a verification field in the host's message against two candidate expected values, one expected value being a keyed hash of the host's recovered message and the other expected value being the bitwise complement of the keyed hash, with the polarity of

the matching candidate determining the active commitment key selection.

[0083] Selection of the active commitment key by cryptographic invariant rather than by plaintext mode indicator provides an additional privacy property. The mode of any specific exchange is part of the protected payload rather than part of the wire-observable framing. An attacker observing the wire cannot determine whether a given exchange used *ira* or *wairua* as the active commitment key without breaking the underlying cryptographic invariant. The same wire-indistinguishability property extends to the Q_HEALTH state-query exchange described above and to every other category of in-session message defined by the protocol: every response shares the same wire format, the same payload size, and the same authenticator construction, so an attacker observing the wire cannot determine the semantic category of a given exchange, whether it is a per-domain attestation, a Q_HEALTH state report, or any other in-session message, without breaking the cryptographic invariant. Message categories are therefore protocol-internal distinctions visible only to endpoints holding the session state, and present no wire-observable side channel by which an attacker could selectively target, filter, or correlate exchanges by semantic type.

6.7 Hardware-Rooted Provable Erasure

[0084] The session-continuity proof property described above admits a converse use of independent commercial and security significance: verifiable proof that a chip's session has ended, and that any cryptographic capability bound to the prior session's *wairua* has been irreversibly lost. The present subsection describes the embodiment of this property as a primitive for hardware-rooted cryptographic erasure.

[0085] In the embodiment, a payload protected by a content-issuing application is encrypted under a content key derived in part from the chip's current-session *wairua*. The derivation is performed by the cascaded HORO-plus-BLAKE3 conditioner described in connection with the attestation mechanism above, taking *wairua*, the protected payload identifier, and any application-supplied freshness as inputs. The chip retains the

derived content key only in volatile registers for the duration of the session; the derived key is not committed to any write-once storage cell, nor to any retention element in the chip's mask design, the chip having no battery, no on-die non-volatile memory other than the write-once *ira* storage, and no path by which *wairua* or any value derived from *wairua* may persist across an interruption of the chip's external power supply.

[0086] Erasure of the protected payload is accomplished by interrupting the chip's external power supply for any duration sufficient to discharge the chip's volatile registers. No protocol-level command, no fresh-session handshake, and no other in-session signal removes *wairua* or any value derived from *wairua*; the only mechanism by which the prior *wairua* can be lost is the physical discharge of the registers that hold it. In every such case, the prior *wairua* is irrecoverably lost and the derived content key is irrecoverably lost with it. The chip's mask design, providing no mechanism by which the prior *wairua* could be preserved, makes the loss of the derived key a physical consequence of the loss of volatile state rather than a software-asserted property; the chip is mechanically incapable of retaining the prior *wairua* across a power interruption regardless of any external instruction the chip might receive, regardless of any modification an attacker might make to the chip's firmware, and regardless of any internal state the chip's finite-state machine might occupy.

[0087] The chip subsequently proves the erasure to the content-issuing application by responding to a session-continuity probe (comprising any deterministic query whose response is a function of *wairua*) and producing a response that does not match the response previously recorded by the application from the prior session. The application's verification proceeds in two parts: an *ira*-keyed attestation confirms that the responding chip is the same physical chip with which the application registered (the chip's permanent identifier being unchanged across the power interruption), and the mismatched session-continuity response confirms that the responding chip's *wairua* has been replaced with a fresh sample (the prior session's volatile state having been lost). The combination of identity-match with session-mismatch is cryptographic proof that the same chip has undergone session-state loss between the two observations.

[0088] The combined proof requires no trust in the chip's firmware, no trust in the chip's manufacturer beyond the manufacturer's role in producing genuine hardware of the published mask design, and no trusted execution environment attestation chain. The erasure is enforced by the physical absence of any retention path in the silicon; the proof is constructed from primitives the chip exposes regardless of firmware state; and the verifier requires only the chip's previously-established public commitment derived at registration.

[0089] The above proof of erasure is conditional on the responding device being genuine hardware of the published mask design at registration time. The condition is satisfied in the present invention not by reliance on a vendor-signed endorsement, which would introduce an upstream issuing authority the chip's design otherwise refuses, but by reliance on the audited manufacturing process by which the chip is fabricated. The chip's mask design and fabrication process are intended for public publication and third-party audit; verification of any specific chip's hardware authenticity is therefore reducible to verification of the chip's provenance thru the audited supply chain. The architecture differs from prior art trusted-platform-module and secure-enclave constructions, in which proof of hardware authenticity reduces to verification of a vendor signature under a vendor endorsement key, said vendor key being a centralized cryptographic root subject to coercion of a single party. In the present invention, no such centralized root exists; the trust anchor is the audited supply chain, and coercion of the trust anchor requires coercion of every audited fabrication step rather than coercion of a single signing authority.

[0090] Embodiments of the hardware-rooted provable-erasure primitive include, without limitation: digital content licensing in which a licensor verifies that a licensee's playback device has irreversibly lost the capability to decrypt licensed content upon end-of-rental; lending of digital library media in which a return event is verified cryptographically rather than asserted by client software; tamper-evident continuous operation of high-assurance systems in which an unexpected loss of session state evidences interruption; secure transfer of capability between devices in which the source device's loss of capability is

verified before the destination device's grant is honoured; and any analogous setting in which a remote party requires cryptographic confirmation that a counterparty's previously-held capability is no longer recoverable.

6.8 The Silent-On-Corruption Failure Mode

[0091] The chip's response to any failure condition is silence at the wire. The wire driver's drive-enable signal is gated off; the wire returns to its pull-down idle; and the chip emits no other signal at the chip's external pin.

[0092] Failure conditions that cause the chip to enter the silent NGARO state are integrity failures of the chip itself, and comprise: a byzantine quorum failure of the chip's permanent-identifier storage, including the cryptographic-agreement failure in which the valid storage pairs do not agree on a single recovered *ira*; a true-random-number-generation health failure; and any non-operational condition of the chip's finite-state machine not enumerated above. The last condition is a catch-all for any internal state of the chip not anticipated as operational by the chip's mask design, including a bus stall, an undefined state code, or a coherency failure in the cryptographic conditioner.

[0093] A cryptographic verification failure on a received protocol message is, by contrast, not an NGARO condition and does not silence the chip. Such a failure indicates ordinary message corruption or a transient desynchronisation of the two endpoints' session state; the chip drops the affected exchange, does not advance its session ratchet, and the host re-attempts the exchange or initiates a fresh session. A verification failure of the per-session secret's own integrity check, distinct from a received-message verification failure, bricks the current session only and is recovered by a power-cycle and a fresh handshake, again without entering NGARO. NGARO is reserved for failures of the chip's permanent integrity, not for the ordinary corruption and resynchronisation events of in-session messaging.

[0094] The chip does not emit an error code, a status byte, a corruption indicator, or

any other distinguishing signal at the wire under any failure condition. Any signal the chip emitted at the wire would be observable by any party with access to the wire. A chip that distinguished its failure modes thru its wire output would make its failure modes observable to parties from whom such information should not be available. An attacker who could distinguish a chip present but with failed byzantine quorum from a chip not present learns information about the chip's storage state; an attacker who could distinguish a chip with failed true-random-number-generation diagnostics from a healthy chip learns information about the chip's entropy source; an attacker who could distinguish a chip in NGARO due to an *ira* cryptographic-agreement failure from a chip not in the system learns information about the chip's storage-integrity state. None of these distinctions assists the chip's owner; all of them assist an attacker.

[0095] The NGARO state collapses all failure conditions to a single externally-observable state: silence. The wire is at the low logic level. Whether the chip is in the ORA state and idle, in the NGARO state, unpowered, or physically removed from the system, the wire is at the low logic level. The chip's owner may distinguish among these conditions by means other than the wire (a power indicator, a physical inspection, or a recovery procedure); an external observer accessing only the wire cannot distinguish among them.

[0096] The implementation of the NGARO state at the wire driver comprises an AND gate computing the drive-enable signal as the logical AND of three signals: a chip-operational status signal, derived combinationally from the byzantine quorum check; a slot ownership indicator; and the endpoint's computed wire output value. When the chip-operational signal is at logic low, the drive-enable signal is at logic low, the wire driver does not drive the wire, and the wire is held at the low logic level by the external pull-down resistor. The mechanism is structurally simple and is enforced by the chip's mask design.

6.9 Inverted-Pair Write-Once Identity Storage

[0097] The chip's permanent identifier is stored in a plurality of write-once storage groups, each group comprising at least two cryptographically-related write-once cells. In some embodiments, each storage group comprises two cells, with the chip's permanent identifier stored in one cell of the group and the bitwise complement of the permanent identifier stored in the paired cell. In some embodiments, four such storage groups are provided.

[0098] The validity check applied to each storage group requires that the bitwise exclusive-or of the cells in the group equals the all-ones value, and that neither cell in the group is at the all-zero state. The first condition confirms that the cells contain cryptographic inverses; it alone already excludes the unprovisioned (both-cells-zero) state, since the all-zero value is not its own complement and an all-zero pair does not satisfy the all-ones exclusive-or. The second condition is therefore not directed at the both-zero case but at the degenerate identifier: a pair in which one cell is all-zero and its complement is all-ones satisfies the exclusive-or check yet encodes a permanent identifier of all-zero (or, symmetrically, all-ones), a structurally trivial value reserved against use; requiring that neither cell be all-zero excludes such a group from contributing a degenerate identifier to the recovery.

[0099] The chip classifies each storage group at reset into one of three categories: a group whose cells are exact bitwise complements is a *burned* group, a relationship the all-zero state cannot satisfy since no value is its own complement, so the burned and blank categories are inherently disjoint; a group whose cells are both at the all-zero state is a *blank* group; and any other group, including one in which a single cell has a stuck or flipped bit, is an *invalid* group counting toward neither category. The chip's provisioning state is then determined by a byzantine quorum over each category independently rather than by a single fragile predicate over the whole array. A quorum of mutually agreeing burned groups places the chip in a provisioned operational state (ORA, HARA, or WHARA according to the count of valid burned groups); a quorum of blank groups

places the chip in the unprovisioned KORE state, ready for the *whakaira*; and a chip reaching neither quorum enters the silent NGARO state.

[0100] The burned-group quorum requires unanimity among the burned groups, not merely a majority. A group that passes the exclusive-or completeness check is internally perfect across its full width; a burned group that is internally perfect yet recovers an identifier different from that of another burned group is therefore not a degraded group but an anomalous one, since ordinary degradation drives a failing group to the invalid category (breaking its exclusive-or relationship) rather than leaving it complete while recovering a second distinct identifier. The only mechanisms that produce a complete-but-disagreeing burned group are deliberate tampering and coordinated fault. Consequently, if any burned group disagrees with any other burned group on the recovered identifier, regardless of the split ratio, including the case of a lone dissenting group against three agreeing groups or two groups against two, the chip enters the NGARO state and is silent. The chip does not resolve such a disagreement by majority count, because the count is meaningful only under an assumption of independent honest failures that a complete-but-disagreeing group violates: an adversary able to forge one coherent burned group may be able to forge several, so the larger faction is not presumptively the genuine one. The chip operates only when every burned group agrees on a single identifier, and otherwise declines to attest under an identity it cannot unambiguously determine.

[0101] Determining the unprovisioned state by a quorum of blank groups, rather than by requiring the entire array to read all-zero, is a deliberate design choice that corrects a fragility otherwise present at the most common shipping state. Were the unprovisioned state defined as the entire array reading all-zero, a single stuck or flipped bit anywhere in the array, arising from a fabrication defect or a single-event upset on an as-yet-unprovisioned chip, would prevent the array from reading all-zero and the chip would fail to be recognised as unprovisioned; lacking any burned quorum, such a chip would enter the silent NGARO state and be indistinguishable from a dead or absent chip, when in fact it is a sound chip awaiting provisioning. The all-zero predicate

has no redundancy: its distance from the not-all-zero condition is a single bit, so the unprovisioned state, which every chip occupies before the *whakaira*, would be the least fault-tolerant classification in the architecture. By giving the blank state the same byzantine quorum treatment as the burned state, a stuck or flipped bit in one or even two groups of a fresh chip leaves a quorum of blank groups intact, and the chip is correctly recognised as unprovisioned and remains provisionable.

[0102] In the single combination in which both a quorum of burned groups and a quorum of blank groups are simultaneously present, namely an embodiment of four groups in which exactly two groups are fully burned and agree on an identifier while the remaining two groups are pristine blank, the provisioned classification takes precedence and the chip operates on the identifier recovered from the burned groups. This precedence is sound because the exclusive-or completeness invariant guarantees that the two burned groups were burned across their full width and therefore recover an identifier of full entropy; a chip exhibiting this combination is one whose *whakaira* wrote two complete groups, and the identifier so recovered is a legitimate, full-entropy identity at the byzantine minimum. The combination is in any case not reachable by the normal operation of the energy-gated burn, which writes all groups to completion or none, and arises only from a hardware fault of the kind described below.

[0103] A naive implementation of redundant identifier storage would store the permanent identifier in a plurality of identical replicate cells and recover it by majority vote among the replicates, applying a single layer of agreement. The present inverted-pair layout is a deliberate departure from that naive implementation and applies two independent layers of checking: a local per-group exclusive-or validity check, admitting a group only if its two cells are exact bitwise complements; and a global byzantine agreement among the groups that survive the local check. The naive majority-vote implementation is retained as an embodiment, but the inverted-pair layout is preferred.

[0104] The exclusive-or validity check is a completeness invariant on the burn, exploiting the failure direction of write-once cells. A write-once cell that has not been

programmed reads as the all-zero state; an incompletely programmed cell, such as one whose programming pulse was too short or whose burn timing was miscalibrated, reads with its unprogrammed bit positions still at zero. For the exclusive-or of a storage group to equal the all-ones value, every bit position must differ between the two cells, requiring at every position that one cell read one and the paired cell read zero. An incomplete burn leaves the same low-order bit positions unprogrammed in both the cell and its complement, so at each such position both cells read zero, the exclusive-or reads zero rather than one at that position, and the all-ones requirement fails. There is no partial-burn outcome that satisfies the all-ones check: because unprogrammed cells fail toward zero rather than toward a chosen value, the all-ones condition is achievable only when the full width of both cells has been programmed to genuine complementary values. The exclusive-or check therefore certifies not merely that the two cells are inverses but that the burn completed across the entire identifier width.

[0105] This property defends against a failure class that majority-vote replication does not catch: a systematic, fleet-wide provisioning error, such as a production burn timer miscalibrated to terminate the programming pulse early, that programs only a fraction of each identifier's bits on every chip of a production run. Under naive majority-vote storage, every chip of such a run would store identifiers that are structurally valid, mutually agreeing, and non-zero, passing every check while silently carrying a fraction of the intended entropy, so that the entire fleet would operate at a reduced effective key size with no failing unit to reveal the error. Under the inverted-pair layout the same error fails the exclusive-or completeness invariant on every chip, because the unprogrammed low-order bits collide at zero in each cell and its complement; the defect is thereby surfaced uniformly across the run rather than concealed. The inverted-pair layout additionally renders the unprovisioned state structurally distinct from any provisioned value rather than distinguishable only by content, and converts the complement cell from mere replication into a per-group integrity code that filters incompletely or incorrectly programmed groups out of the recovery before the byzantine vote is taken.

[0106] The byzantine quorum threshold in some embodiments is two valid storage

groups out of a total of four storage groups, with all valid storage groups required to agree on the recovered identifier. In some embodiments, the threshold may be configured to other values based on the chip's intended deployment population and acceptable failure probability. The thresholds may be derived from population failure-rate calculations, including the calculation that, for an embodiment with four storage groups and a quorum of two, a per-pair failure probability of approximately 0.79 percent yields a whole-chip failure probability below 10^{-15}, which is suitable for population-scale deployment.

[0107] The inverted-pair storage layout absorbs several classes of corruption below the quorum-failure boundary. A single-event upset, such as a cosmic-ray-induced bit flip in one cell of one storage group, invalidates that group thru the validity check, leaving the remaining groups operational; if three groups remain valid, the chip operates in the HARA state. A latent manufacturing defect that manifests after provisioning, such as a marginal eFuse cell or a cell with reduced retention, appears as a single-group invalidity and is treated identically to a single-event upset. An electromagnetic disturbance of sufficient magnitude to flip bits in one or more cells invalidates the affected groups; the remaining valid groups continue to support chip operation. A partial-burn outcome, in which some storage groups are validly burned and others remain kore, is not a normal outcome of the *whakaira*: the energy-buffer threshold gate and supply isolation guarantee the burn runs to completion against external power interruption, so a partial burn arises only from a hardware fault outside the engineering margin, such as a failed or cracked burn capacitor, and is correspondingly rare. If it does arise, the chip applies no special-case rejection; the partial-burn array is classified by the same byzantine quorum logic as any other corruption. If the count of validly burned groups that agree on a single recovered *ira* meets quorum, the chip operates in the corresponding degraded state (HARA or WHARA) on the identifier those groups recover; if the count falls below quorum, or if the valid groups do not agree on a single *ira*, the chip enters the NGARO state and is silent. The exclusive-or completeness invariant guarantees that any group counted as validly burned was burned across its full width, so a quorum of burned

groups recovers an *ira* of full entropy and no separate commit indicator is required to certify that the burn completed.

[0108] The integrity verification of the chip's permanent identifier is performed exclusively at read time by the byzantine quorum check, and no per-cell read-back verify is performed during the burn sub-step. This architectural choice is intentional. A per-burn read-back verify implementation would detect a defective storage cell at the moment of its attempted write and, having no recourse against a write-once cell that has failed to take, would necessarily abort the provisioning ceremony and place the chip in a permanently inoperable state on the first encounter with any cell-level defect. The chip's owner would observe such a chip as bricked from manufacturing-process variation that is not in fact systemic, the defect being statistically present in any large population of fabricated devices at the cell level even when no group of cells is jointly compromised. The byzantine quorum check, by contrast, operates over the full set of storage groups simultaneously and applies the failure threshold to the count of compromised groups rather than to the existence of any single compromised cell. A chip with a defective cell in storage group 0, when verified at read time, presents an invalidated group 0 and (assuming the remaining three groups burned correctly) a valid quorum of three groups; the chip is reported in the HARA state and is fully operational. The same chip, verified instead by per-burn read-back during TAHU, would have its ceremony aborted at the first slot and the chip would be permanently NGARO. Deferring verification from write time to read time therefore converts a class of manufacturing defects from chip-level failures into silently-absorbed group-level failures, increasing the yield of usable chips against the same defect rate. The architecture is intentionally more permissive at burn time and more rigorous at read time, the burn-time circuit having visibility only into the single cell being written while the read-time circuit has visibility into the entire storage array and may apply byzantine reasoning across the redundancy.

[0109] No external signal sequence at the chip's external port can alter, replace, or extract the contents of the write-once storage cells. The cells are coupled to internal silicon paths and to the chip's finite-state machine, but not to the external port in any

path that exposes the cells' contents in plaintext or in any reversible encoding. The chip's permanent identifier is therefore non-destructively inviolable from outside, in the sense that an attacker possessing the chip and operating the chip's external interface has no means by which to extract, substitute, or modify the chip's permanent identifier. The permanent identifier participates in cryptographic operations inside the chip only as a key for cryptographic commitments; the commitments emitted at the chip's external port are functions of the identifier under preimage-resistant cryptographic hash, and an external party in possession of any quantity of such commitments cannot recover the identifier itself by any non-destructive computational means.

[0110] Recovery of the chip's permanent identifier by an attacker would require physical invasive analysis of the chip's silicon, including but not limited to: decapping of the chip package and layer-by-layer mechanical or chemical removal of the chip's metal layers to expose the write-once storage cells for optical or electron-microscope readout; focused ion beam milling to probe internal silicon nets; laminographic X-ray imaging of the chip's internal structure; or other destructive techniques. The chip's mask design may, in some embodiments, further frustrate such physical invasive analysis thru one or more anti-invasive features, including: a top-metal active mesh covering the write-once storage cells and the cryptographic conditioner, the mesh configured to detect penetration and cause the chip to enter the failure-response state described above; one or more passive shielding layers comprising metallization patterns configured to obstruct laminographic X-ray imaging of the chip's internal structure; an irradiation-detection circuit configured to cause the chip to enter the failure-response state upon detection of an attempt at invasive analysis using charged-particle or photon-based probes; one or more dummy circuit blocks resembling but not implementing the cryptographic functions of the chip, the dummy circuits configured to mislead reverse-engineering effort; and a tamper-evident package coating configured to render decapping observable to the chip's owner. The choice and combination of anti-invasive features is a matter of implementation; the protocol described in the present specification is independent of the chosen anti-invasive features and is enforced at the silicon level regardless of which

combination is selected.

6.10 Operating-Envelope Enforcement

[0111] The anti-invasive features described above defend against physical intrusion into a powered-down or probed chip. A complementary class of defense, described in the present subsection, protects the chip while it is operating normally against fault-injection attacks that do not require physical intrusion but instead perturb the chip's operating conditions to corrupt the evaluation of its own logic. The chip's correctness guarantees, including the byzantine quorum classification, the finite-state machine's forward-only transitions, and the energy-gate and burn sequencing, all presuppose that the chip's combinational and sequential logic evaluate correctly; that presupposition holds only while the chip operates within a bounded envelope of physical conditions. An adversary who drives the chip outside that envelope, for example by briefly collapsing or overshooting the supply voltage at a chosen instant, can cause a comparison to mis-evaluate, an instruction or state transition to be skipped, or a stored value to be mis-sensed, and thereby subvert a guarantee that is otherwise sound.

[0112] The chip therefore includes, in some embodiments, one or more operating-envelope sensors and a monitor that holds the chip inoperative and wire-silent whenever any monitored condition is outside its valid range. The monitored conditions comprise, in various embodiments, any one or more of: supply voltage below an undervoltage threshold or above an overvoltage threshold; a rate of change of supply voltage exceeding a threshold, such a transient being a signature of a glitch attack even when the absolute voltage remains within bounds; die temperature below a low-temperature bound or above a high-temperature bound, low-temperature operation being associated with data-remanence and metastability-manipulation attacks and high-temperature operation with retention failures and entropy-source bias; a rate of change of die temperature exceeding a threshold; an internal clock frequency, derived from the chip's free-running ring oscillator, outside a valid band, an out-of-band frequency indicating that an attacker is manipulating the oscillator thru

voltage or temperature to push the finite-state machine out of its timing margin; an ambient electromagnetic field strength exceeding a threshold, a strong field being the vector of electromagnetic fault injection; incident light or charged-particle flux exceeding a threshold, indicating an optical or particle-beam fault-injection attempt against a decapsulated die; a static or low-frequency magnetic field strength or magnetic anomaly outside a valid range, as sensed by a magnetometer, distinct from the radio-frequency electromagnetic sensor and directed at magnetic fault injection; a mechanical shock, vibration, or acceleration exceeding a threshold, as sensed by an accelerometer, indicating mechanical fault injection or the handling associated with mounting a die in a probe station; an ambient pressure outside a valid range, indicating decapsulation, evacuation, or other package-integrity compromise; a chemical signature indicating the presence of an etchant, solvent, or other agent associated with chemical decapsulation; and any further physical invariant whose departure from its valid range is consistent with a fault-injection or invasive-analysis attempt against the chip.

[0113] When any monitored condition is outside its valid range, the monitor holds the chip's finite-state machine inoperative and gates off the wire driver, so that the chip neither performs identity operations nor drives the channel, its wire behaviour during the out-of-envelope condition being indistinguishable from that of an unpowered or absent chip in the manner of the silent failure mode described above. The out-of-envelope condition is latched: the monitor does not resume operation the instant the monitored conditions return within range, because an attacker able to drive a condition out of range and back can otherwise ride the boundary and catch the finite-state machine mid-recovery in a corrupted state. The latch is cleared only by a complete power-on reset, recognised by the chip's supply falling below a hold-off voltage low enough to guarantee that the chip's volatile state has been fully discharged before operation re-arms. Operation therefore resumes only after all monitored conditions are within their valid ranges and the chip has undergone a genuine power cycle, whereupon the storage quorum is re-evaluated from a clean state in the ordinary way. This latch-and-full-cycle discipline ensures that the only exit from a tripped envelope is a clean restart, never

a partial recovery into a state an attacker timed. The operating-envelope monitor is the general form of the burn-specific energy-buffer threshold gate described above: the energy gate is the application of the undervoltage invariant to the single TAHU burn, whereas the operating-envelope monitor applies the full set of environmental invariants to all operation of the chip. The two are complementary and may be present together. The choice and combination of operating-envelope sensors is a matter of implementation, and the protocol described in the present specification is enforced at the silicon level regardless of which combination is selected.

6.11 Atomic Provisioning Commit

[0114] The provisioning ceremony of the present invention is triggered by the chip's owner in the field, at a moment of the owner's choosing, on whatever bench or installation environment the owner happens to be using. The chip is not provisioned in a controlled manufacturing facility under the supervision of the chip's manufacturer; the chip is not binned out before shipment if its provisioning is incomplete; no operator-side recovery procedure may be relied upon to complete a partially performed provisioning. The chip must therefore present only valid named states of its finite-state machine at every instant during and after the provisioning ceremony, under arbitrary timing of external power interruption. This requirement is, in the present specification, the atomic provisioning commit property; it distinguishes the present invention from prior art identity-creation arrangements that achieve atomicity by performing the identity-creation event at the manufacturer's facility and that re-introduce the vendor trust path the present invention rejects.

[0115] The atomic provisioning commit property is implemented by an energy-budgeted burn-initiation gate. The chip comprises an energy storage element, a sensor measuring stored energy in the element against a configured threshold, and a finite-state machine that refuses to begin the burn of the chip's write-once storage cells until the sensor reports that the threshold is met. The threshold is configured to a stored-energy quantity sufficient, with engineering margin, to drive the entire burn sequence to completion

at the required burn voltage. Once the burn begins, the energy storage element is electrically isolated from the chip's external supply by a switching element coupled between them, and the burn proceeds to completion under power supplied by the storage element regardless of any subsequent interruption of the chip's external power supply. Interruption of the external supply before the threshold is met leaves no storage cell written and the chip presents the KORE state on the next power-on cycle. Interruption after the burn begins is irrelevant because the burn is no longer dependent on the external supply.

[0116] The entropy-gathering sub-step (KOHI), the chaotic-dynamical amplification sub-step (HORO), the cryptographic-conditioning sub-step (BLAKE3), and the randomized pre-burn delay sub-step (TATARI) of the provisioning ceremony precede the burn-initiation gate and operate on volatile registers internal to the chip. No write-once storage cell is touched during these sub-steps. Interruption of the chip's external power supply during any of these sub-steps drops the volatile registers and the chip presents the KORE state at next reset; the conditioned identifier candidate that existed in the volatile registers at the moment of interruption is lost and is not recoverable by any party, and the chip's owner may re-attempt the ceremony, the re-attempt drawing a fresh sample of physical entropy at the moment of re-attempt and producing a different conditioned identifier. An adversary observing or interrupting any number of failed attempts therefore learns no information about the identifier eventually committed, because each successful commit binds to fresh entropy drawn at the moment of the successful attempt, said entropy being unrelated to the entropy of any prior interrupted attempt.

[0117] In a non-limiting embodiment, the chip's write-once storage cells comprise spin-transfer-torque magnetic-tunnel-junction (STT-MRAM) cells configured for parallel write. The complete burn of the inverted-pair storage array described in connection with FIG. 6 (eight 256-bit slots) requires approximately two nanojoules at approximately one volt over a duration of approximately one hundred nanoseconds. The energy storage element required is correspondingly modest; in a representative embodiment, the chip's

intrinsic on-die and on-package power-supply decoupling capacitance, comprising on the order of ten nanofarads in typical packaging, exceeds the burn-energy budget by several orders of magnitude and itself serves as the energy storage element with no dedicated external component required. The energy sensor in this embodiment comprises a comparator measuring the chip's internal supply rail voltage against a reference, the comparator asserting the burn-permission signal when the rail voltage exceeds the threshold consistent with sufficient stored energy.

[0118] In a further non-limiting embodiment, the chip's write-once storage cells comprise electrically-programmable fuses whose burn voltage and energy requirements exceed the budget of intrinsic on-package decoupling. In this embodiment, the energy storage element comprises an external capacitor coupled to the chip's external supply pin (or to a dedicated burn-rail pin), the capacitor's capacitance selected to store, with margin, the energy required to burn the complete inverted-pair storage array at the eFuse burn voltage. In a representative embodiment of this kind, an external capacitor on the order of one millifarad coupled in conjunction with an on-die charge pump producing the eFuse burn voltage is sufficient. The energy sensor in this embodiment comprises a comparator measuring the external capacitor's terminal voltage against a reference. The switching element comprises a transistor or schottky diode coupled between the external supply and the capacitor, configured to disconnect the capacitor from the external supply at the moment burn begins.

[0119] In further non-limiting embodiments, the chip's write-once storage cells comprise embedded floating-gate cells, antifuses, mask-programmed read-only memory cells, polymer write-once cells, or any combination of the foregoing; the energy storage element comprises any combination of on-die capacitance, on-package capacitance, and external capacitance; and the energy sensor comprises any analog circuit measuring stored charge or rail voltage and emitting a single binary burn-permission signal. The atomic provisioning commit property is independent of the specific technology selected for the storage cells, the specific physical embodiment of the energy storage element, or the specific implementation of the sensor; the requirement is that the

chip's finite-state machine refuse to begin the burn until the sensor confirms sufficient stored energy, and that the burn, once begun, complete under power supplied by the storage element rather than by the external supply.

[0120] Failure modes of the energy-buffered burn are absorbed by the same KORE-or-ORA two-state invariant. A chip whose energy sensor never reports the threshold met (because the external supply is insufficient to charge the storage element to threshold, or because the sensor is defective) never enters the burn state and therefore never writes any storage cell; the chip presents the KORE state indefinitely until a sufficient external supply is provided or the chip is replaced. A chip whose burn begins but whose energy storage element fails mid-burn (a fault outside the design margin, such as a cracked capacitor) leaves a partially-burned storage array; the chip's classifier behaviour with respect to a partially-burned array is determined by the byzantine quorum logic described in the preceding subsection, and depending on the extent of partial burn the chip may present any of the operational states (ORA, HARA, WHARA, or NGARO) on the next power-on cycle. This residual hazard is bounded by the engineering margin of the energy budget and is not, in any case, exploitable by an external attacker thru normal power-supply manipulation.

[0121] The atomic-commit property guaranteed by the energy-buffered burn architecture is specific and should not be overstated. What it guarantees is that no persistent *intermediate finite-state-machine* state between KORE and ORA is reachable by interruption of the chip's external power supply: the pre-burn sub-steps are volatile and revert to KORE, and the TAHU burn, once begun, runs to completion on the isolated energy buffer regardless of supply manipulation. The energy gate therefore removes power interruption as a means of leaving the finite-state machine partway between unprovisioned and provisioned.

[0122] What the energy gate does not and cannot guarantee is that the committed result is necessarily ORA. The committed state recovered at the next reset is whatever the byzantine quorum classifier reports over the cells as actually written, and a chip may come up at KORE, ORA, HARA, WHARA, or NGARO directly out of provisioning. A

fault injected into the storage cells during the TAHU burn, by a timed electromagnetic, optical, or particle disturbance of the kind the TATARI randomized delay and the operating-envelope monitor defend against, or a latent cell defect that causes a pair to burn incompletely, can corrupt one or more pairs as they are written, so that the chip is provisioned directly into a degraded operational state (HARA or WHARA) or, if the corruption drops the result below quorum or produces a disagreeing pair, directly into NGARO. The degradation states are thus reachable at the moment of provisioning, not only on the subsequent time scales of hours to years over which single-event upset, retention drift, or in-field disturbance corrupt the cells of an already-provisioned chip.

[0123] The provisioning ceremony and the moments immediately surrounding power-on are, for this reason, the chip's highest-exposure window rather than a uniquely protected one: the entropy is live in volatile registers, the cells are being written, and the classifier is first evaluating. The architecture's defenses are concentrated accordingly, and are layered rather than singular: the energy gate and supply isolation defeat power-interruption attacks on the burn; the TATARI randomized delay denies the timing required by an injection attack aligned to the burn instant; the operating-envelope monitor holds the chip inoperative and silent if voltage, temperature, clock, electromagnetic field, or radiation leaves the valid range during the ceremony; and the byzantine quorum classifier, with its unanimity requirement, catches at read time any corruption that nonetheless reaches the cells, surfacing it as a degraded state or as silence rather than as a confidently-attested wrong identity. The atomic-commit property is enforced by silicon physics and the energy budget rather than by software, and is independent of the chip's firmware, the chip's host endpoint, and any operator-side recovery procedure.

6.12 Application Surface of the Device-Identity Primitive

[0124] The device-identity primitive provided by the present invention is application-agnostic. The chip's identity is created in the chip's own silicon; the chip's identity is expressed only thru domain-separated commitments; and the chip's

identity is owned by the chip. What the device-identity primitive is used for downstream is a matter outside the chip's concern and outside the scope of the present claims. The breadth of the application surface is the breadth of the market for the primitive.

[0125] In an Internet-of-Things device identity application, a device originally provisioned with a per-device certificate signed by the device vendor or by a third-party identity service is replaced by a device provisioned in accordance with the present invention. The third-party identity service or device vendor is removed from the trust path; the device's identity is created in the device's own silicon by the device's owner (which may be the original equipment manufacturer integrating the chip, or the end user if the original equipment manufacturer defers provisioning to the end user); and the Internet-of-Things platform receives the device's public commitment at registration time and verifies subsequent attestations without further third-party involvement.

[0126] In a hardware-wallet application, a cryptographic wallet seed is derived from the chip's permanent identifier thru one or more domain-separated derivations, eliminating any factory-injected key from the wallet's seed chain.

[0127] In a user-sovereign identity binding application as described in the related TOKEN application, the chip's permanent identifier serves as the device-side input to a device-to-user binding derivation, with the user's TOKEN handle serving as the user-side input. The binding derivation is the subject of the TOKEN application and is outside the scope of the present claims.

[0128] In an automotive secure boot application, a vehicle's electronic control unit is provisioned with a chip configured in accordance with the present invention; the chip's identity survives ownership transfer of the vehicle without re-provisioning, because the chip's identity is owned by the chip rather than by the vehicle's current owner.

[0129] In a medical device patient-private binding application, a medical device holding patient-private data is provisioned with a chip configured in accordance with the present invention by the patient or by the patient's care provider, eliminating any device-manufacturer involvement in the chip's data-stream identity. In a sharper

application of the hardware-rooted provable-erasure primitive, a patient's record, or a selected portion thereof, is encrypted under a content key derived from the device's per-session secret *wairua*, so that a care provider's machine can decrypt and display the record only while its session is live; a power-off or reset of the machine destroys the session *wairua* and with it the decryption capability, and the patient, or a remote party acting for the patient, can verify the revocation by the same identity-match-with-session-mismatch proof described above, confirming that the same machine no longer holds the capability to decrypt the record. The patient may grant access to distinct portions of the record under distinct session-derived keys and revoke each portion independently and provably. This protects the machine-recoverable copy of the record; it does not and cannot prevent a viewer from photographing a displayed screen, transcribing, or otherwise capturing the information by means outside the cryptographic system, and the property claimed is the provable revocation of the machine's capacity to decrypt the stored record, not the prevention of all human recollection or out-of-band capture.

[0130] In an industrial control attestation application, an industrial control system is provisioned with chips configured in accordance with the present invention by the operating entity (a utility, a refinery, a power plant, or another operator skeptical of vendor trust), eliminating any device-vendor certificate authority from the trust path.

[0131] In a defense supply-chain post-fabrication substitution detection application, a chip configured in accordance with the present invention is provisioned by the defense supplier at a known time and place; any subsequent substitution of the chip would require the substituted chip to possess a permanent identifier created in the original chip's silicon, which is information-theoretically impossible because the original chip's identity was created in silicon-native entropy that no other chip can replicate.

6.13 Experimental Validation

[0132] The protocol described in the present specification has been implemented and validated on hardware. A non-limiting example implementation comprises a Lattice ECP5-25F field-programmable gate array configured to perform the chip-side protocol, coupled by a single conductor to a Raspberry Pi RP2040 microcontroller configured to perform the host-side protocol. A nine-test simulation testbench exercises the chip-side finite-state machine thru the provisioning ceremony, two repeated permanent-mode authenticated exchanges (to demonstrate determinism within a session), a session-mode authenticated exchange (to demonstrate mode separation), a state-query authenticated exchange (to exercise the Q_HEALTH state-vector reporting embodiment), and two single-bit-fault exchanges (to exercise the silent-on-corruption failure mode). All nine simulation tests passed without modification of the protocol implementation. A twelve-test host-side implementation crate exercises the host-side protocol and passes all twelve tests, including end-to-end exchange of the canonical test vector thru an in-process chip emulator. The canonical test vector employed throughout the host-side suite is the keyless BLAKE3 digest of a fixed ASCII preimage. The preimage begins with the phrase “I had the winning ticket but forgot the number. How can I prove it won?” concatenated with a short withheld suffix; the full preimage evaluates to `1a6c0220d754bdb8b1685b162923aadd913019372c255ed7e109016f3131f992`, a fixed value chosen so that the byte-order and encoding convention of the cryptographic hash is reproducible across the chip-side and host-side implementations from a single fixed preimage. The chip-side cryptographic hash computation has been independently validated by cross-comparison against the host-side cryptographic hash computation on real silicon: a fingerprint of the chip’s session pad observed on the field-programmable gate array’s output display was matched exactly against the host-side computation, confirming that the byte-order convention of the cryptographic hash is independently validated on real silicon rather than only being self-consistent within a simulator.

[0133] Two distinct permanent-identifier values were generated by the field-programmable gate array on two power cycles during the validation session,

each value produced by the field-programmable gate array's on-die true-random-number generator passed thru the on-die cryptographic conditioner during the chip-internal provisioning ceremony. The two values are cryptographically independent of each other and represent the chip's own entropy rather than entropy injected from outside the chip. The mirror-XOR line code was previously validated bit-exact in a prior validation session: 256 of 256 bidirectional round-trip exchanges completed without bit error over the single conductor between the field-programmable gate array and the microcontroller.

[0134] A fault-injection bitstream that bakes a single storage pair as burned and three storage pairs as kore was loaded onto the field-programmable gate array to simulate an interrupted provisioning. The chip's classifier identified the state correctly: chip_state reported NGARO, wire output was gated off, the LED indicator confirmed the failure mode by a specific blink pattern, and the mirror-XOR heartbeat was absent from the wire. The byzantine quorum was below threshold; the chip refused to operate; the wire was silent. The observed behaviour matches the NGARO state property recited in Claim 8.

[0135] A Computer Program Listing Appendix is submitted herewith pursuant to 37 CFR §1.96 and is hereby incorporated by reference into the present specification. The appendix comprises hardware-description-language source listings for the chip-side embodiment validated in the foregoing experimental validation, including the mirror-XOR encode and decode logic of Claim 3, the leading-edge framing and slot-counting decode of Claim 4, the inverted-pair write-once storage and byzantine quorum classifier of Claim 9, the provisioning ceremony finite-state machine of Claim 1, the challenge-response finite-state machine implementing the per-message wire format of Claim 5 and the handshake-as-entropy-request behaviour of Claim 6, and the on-chip noise-capture circuitry and ring-health classifier supporting the entropy-source health requirement of Claim 8. Standard cryptographic primitives invoked by the listings (including the BLAKE3 hash function) are referenced by their published specifications and represented in the listings as port-interface module declarations rather than as full implementations, the full implementations being publicly available reference code not material to the novelty of the present invention. The listings are exemplary embodiments

and are not intended to limit the scope of any claim to the specific signal names, parameter values, bit widths, structural arrangements, technology targets, or synthesis tool flows depicted therein; a person of ordinary skill in the art will recognise that equivalent embodiments may be realised in alternative hardware-description languages, on alternative silicon technologies, and with alternative module decompositions without departing from the scope of the claims.

6.14 Implementation Variations

[0136] Validation on field-programmable gate array fabric, as described in the preceding section, employs implementation choices that are strictly easier problems on production application-specific integrated circuit silicon. The validation evidence is therefore a lower bound on what production silicon may achieve.

[0137] With respect to true-random-number generation primitives, the field-programmable gate array implementation uses ring-oscillator primitives only. Field-programmable gate array fabric does not natively support the metastability-based, transition-effect-ring-oscillator-based, or subthreshold-inverter-shot-noise-based primitives that production complementary metal-oxide semiconductor silicon supports. Production silicon embodiments may include any combination of the foregoing primitives in addition to ring oscillators, providing additional physics-grade entropy sources beyond what the field-programmable gate array can exercise.

[0138] With respect to write-once storage, the field-programmable gate array implementation emulates the inverted-pair storage layout using bitstream-initialized registers. Production silicon embodiments employ literal write-once storage cells, such as electrically programmable fuses, antifuses, mask read-only memory, or other write-once technologies, and provide electrical irreversibility by physical mechanism rather than by policy.

[0139] With respect to wire input/output, the field-programmable gate array development arrangement uses the field-programmable gate array’s general-purpose

input/output with internal weak pull-down as an approximation to the asymmetric drive arrangement. Production silicon embodiments provide a dedicated pin with a properly sized analog pull-down resistor.

[0140] With respect to slot rate, the field-programmable gate array implementation operates at a slot rate of approximately 50 kilohertz, this rate being the highest rate at which the microcontroller's clock and data recovery routine reliably locks. Production silicon embodiments include dedicated physical-layer logic with no equivalent slot-rate ceiling.

[0141] With respect to power gating of true-random-number generation primitives, field-programmable gate array fabric does not provide block-level power gating. Production silicon embodiments power-gate the true-random-number generation primitive domain, eliminating static power leakage and side-channel emanation of the primitive's oscillation frequency between entropy events.

7 CLAIMS

What is claimed:

1. A method of cryptographic device identity creation, comprising:

receiving, at an external interface of an integrated-circuit chip, an owner-initiated provisioning request signal;

drawing, in response to the provisioning request signal, a quantity of physical entropy by sampling, with one or more noise-capture circuits resident on the chip's silicon, transient physical phenomena present at and within the chip's silicon at the moment of the receiving of the provisioning request signal;

processing the physical entropy thru one or more cryptographic conditioners internal to the chip to produce a cryptographic-quality permanent identifier;

writing the permanent identifier to one or more write-once storage cells inside the chip before any external interface of the chip is capable of transmitting the permanent identifier;

wherein the permanent identifier is not transmitted outside the chip, in plaintext or in any reversible encoding, at any point during or after the provisioning;

wherein the provisioning is initiated solely by an owner of the chip, and no third party, chip vendor, certificate authority, or provisioning service participates in, authorizes, witnesses, or signs the provisioning;

wherein no record of the permanent identifier is created or transmitted outside the chip during the provisioning; and

wherein cryptographic operations involving the permanent identifier subsequent to the provisioning produce only commitments, signatures, or domain-separated derivatives of the permanent identifier, and never the permanent identifier itself.

2. A method of device attestation, comprising:

receiving, at an external interface of an integrated-circuit chip having a permanent identifier produced according to claim 1, a query from a requesting application, the query comprising a domain identifier supplied by the requesting application;

producing, in response to the query, a cryptographic attestation derived as a keyed cryptographic commitment over the chip's permanent identifier and over the domain identifier supplied by the query;

transmitting the cryptographic attestation as the chip's response to the query;

wherein two queries supplying distinct domain identifiers produce cryptographically independent attestations whose correlation requires a preimage attack on the keyed cryptographic commitment function;

wherein the chip's permanent identifier is not exposed in the cryptographic attestation; and

wherein cross-application correlation of the chip across a plurality of requesting applications is not structurally enabled by the chip's architecture, in that such correlation requires the chip's voluntary participation in producing attestations for both correlating domain identifiers, which participation requires user-mediated consent.

3. A method of bidirectional digital communication between an integrated-circuit chip and a host endpoint over a single electrical conductor, comprising:

encoding, at each of the chip and the host endpoint, each transmitted bit as the exclusive-or of an endpoint-specific idle bias value, a bit value to be transmitted, and a bit value most recently decoded by the endpoint from the other endpoint's transmission, wherein the chip uses an idle bias value of binary 1 and the host endpoint uses an idle bias value of binary 0;

decoding, at each of the chip and the host endpoint, each received bit as the exclusive-or of an observed wire value at a sample point of the received bit, the other endpoint's idle bias value, and a bit value the endpoint most recently transmitted;

driving the single electrical conductor to a high logic level during a transmission slot of each endpoint when the encoded bit value of the endpoint evaluates to binary 1; and releasing the single electrical conductor to a high-impedance state otherwise;

wherein one or more pull-down elements coupled to the single electrical conductor, including in some embodiments a respective pull-down element at each endpoint, establish a low logic level at the wire as the wire's idle electrical state;

wherein direct-current balance over the single electrical conductor emerges from the encoding without dedicated balance overhead bits; and

wherein clock recovery at a receiving endpoint is performed by detecting transitions in the wire's state, without dedicated clock recovery hardware beyond a single logic gate.

4. A method of message framing in a single-wire bidirectional protocol, comprising:

transmitting, at the beginning of every message from each endpoint of the protocol, a single binary 1 bit as a leading frame bit, the leading frame bit being transmitted as a transition of the wire's state out of an idle state into active transmission;

transmitting, immediately following the leading frame bit, an N-bit message payload of fixed or otherwise known length N;

ceasing active transmission after the last actively driven binary 1 bit of the payload, the wire returning to its idle state by passive biasing, such that trailing binary 0 bits of the payload are not actively driven onto the wire;

reconstructing, at a receiving endpoint, the trailing binary 0 bits of the payload by slot-counting N slots after the leading frame bit and decoding each non-actively-driven slot as binary 0;

wherein the framing cost is exactly one channel slot per message irrespective of payload length, and the trailing binary 0 bits of the payload are reconstructed by the receiving endpoint without being signalled; and

wherein the maximum total channel slots transmitted per message is the payload length

plus one, the framing never introducing a second dedicated framing slot, and the decode operation is a slot-counted shift register requiring no framing-symbol detection logic.

5. A method of cryptographic message formatting in a single-wire protocol, comprising:

constructing an authenticated message comprising a leading framing bit followed by a payload comprising a masked-message field and a masked-commitment field in either order by a convention shared between the endpoints, the masked-message field being a payload value combined by exclusive-or with a first session pad derived from a session-shared secret, the masked-commitment field being a keyed cryptographic hash of the payload value combined by exclusive-or with a second session pad derived from the session-shared secret, the masked-commitment field being of fixed width equal to the hash output width and the masked-message field being of a payload-determined length, the message carrying no trailing termination marker, no length prefix, and no dedicated framing symbol beyond the leading framing bit;

transmitting the authenticated message over the single-wire protocol of claim 3;

wherein the masked-message and masked-commitment fields together simultaneously provide (a) confidentiality of the payload value, recoverable only by a receiver holding the first session pad; (b) decryption authority, the payload value being recovered by exclusive-or of the masked-message field with the first session pad; (c) message authentication, the receiver recomputing the keyed cryptographic hash of the recovered payload value and verifying it against the masked-commitment field after removal of the second session pad; (d) advancement of the session ratchet, the first and second session pads for the next exchange being derived from the keyed cryptographic hash of the present payload value such that each exchange consumes fresh pads and prior pads are not reused, providing forward secrecy of prior exchanges; and (e) automatic direct-current balance over the channel thru the statistical uniformity of the masked fields, each being the exclusive-or of a value with a cryptographic-quality pad; and

wherein the five protocol functions identified in the preceding clause are delivered by the masked fields of the authenticated message rather than by independently transmitted

dedicated fields, and wherein the construction relies only on symmetric and keyed-hash primitives and not on the hardness of any discrete-logarithm or integer-factorisation problem.

6. A method of cryptographic session initiation between a host endpoint and an integrated-circuit chip, comprising:

receiving, at an external interface of the chip, a single bit transmitted from the host endpoint;

interpreting the single bit simultaneously as a session-initiation request and as a true-random-number-generation request;

drawing, in response to the single bit, a quantity of cryptographic-quality entropy by sampling, with one or more noise-capture circuits resident on the chip’s silicon, transient physical phenomena present at and within the chip’s silicon at the moment of the receiving of the single bit;

emitting the cryptographic-quality entropy as the chip’s response to the single bit;

wherein no separate request-entropy command exists in the protocol, the handshake bit invariably producing fresh entropy as its response.

7. A method of cryptographic identity expression in an integrated-circuit chip, comprising:

maintaining inside the chip a permanent identifier generated by the provisioning method of claim 1 and stored in write-once storage cells of the chip;

establishing, once per power-on cycle of the chip at the first handshake following reset, a per-session secret identifier from fresh physical entropy according to the method of claim 6, the per-session secret identifier residing in volatile registers of the chip, persisting unchanged for the power-on lifetime of the chip, and being irrecoverably lost upon interruption of the chip’s power supply and not upon any protocol-defined event;

receiving, at an external interface of the chip, an authenticated message from a host endpoint;

selecting, in response to a cryptographic invariant carried by the authenticated message, an active commitment key from among the permanent identifier and the per-session secret identifier; and

producing a response to the authenticated message as a keyed cryptographic commitment under the selected active commitment key;

wherein the selection of the active commitment key is conveyed by the cryptographic invariant rather than by any plaintext mode indicator in the authenticated message.

8. A method of failure response in an integrated-circuit chip, comprising:

monitoring, by combinational logic of the chip, for one or more chip-integrity failure conditions including: failure of the chip’s permanent-identifier storage to satisfy a byzantine quorum check, including a cryptographic-agreement failure in which the valid storage pairs do not agree on a single recovered permanent identifier; failure of the chip’s true-random-number-generation primitives to satisfy on-chip health diagnostics; and any non-operational condition of the chip’s finite-state machine; a failure of a cryptographic verification check on a received protocol message being expressly excluded from the failure conditions of this clause and being instead handled by dropping the affected exchange;

gating off, upon detection of any failure condition, a drive-enable signal of the chip’s wire driver such that an external wire interface of the chip is rendered high-impedance for a duration of the failure condition;

allowing the wire’s idle electrical state to be established by one or more pull-down elements coupled to the wire, such that the wire’s behaviour during the failure condition is indistinguishable from the wire’s behaviour when the chip is not physically present in the system;

wherein no error code, status flag, or distinguishing signal other than the wire’s pull-down idle is emitted at the wire under any failure condition.

9. A method of cryptographic identity storage in an integrated-circuit chip, comprising:

allocating within the chip a plurality of write-once storage groups, each storage group comprising at least two cryptographically-related write-once cells;

storing the chip's permanent identifier across the storage groups, wherein storing the permanent identifier in each storage group comprises storing the permanent identifier in one cell of the storage group and storing a bitwise complement of the permanent identifier in a paired cell of the storage group;

classifying, at chip reset, each storage group into one of three categories: a burned group, whose two cells are exact bitwise complements and not both all-zero; a blank group, whose two cells are both at the all-zero state; and an invalid group, being any other group, including a group in which a single cell has a stuck or flipped bit;

performing, at chip reset and optionally during scheduled storage-integrity audits, a byzantine quorum check over the burned groups, requiring agreement among a configured minimum number of burned groups on a single recovered identifier, and a separate byzantine quorum check over the blank groups;

recovering the chip's permanent identifier from the contents of a burned group and entering a provisioned operational state when the burned-group quorum check passes; entering the unprovisioned state when the blank-group quorum check passes; and entering the failure-response state of claim 8 when neither quorum check passes;

wherein the burned-group quorum check passes only if every burned group agrees on a single recovered identifier, such that if any burned group recovers an identifier different from that of any other burned group, regardless of the numbers of groups on each side of the disagreement and without resolving the disagreement by majority count, the quorum check fails and the chip enters the failure-response state of claim 8, the chip declining to attest under an ambiguous identity; a burned group that is internally complete yet disagrees being treated as evidence of tampering or coordinated fault rather than of ordinary degradation, which instead renders a failing group invalid;

wherein, when both quorum checks pass simultaneously, the provisioned operational state takes precedence and the chip operates on the identifier recovered from the burned groups;

wherein recognising the unprovisioned state by a quorum of blank groups, rather than by requiring the entire storage array to read all-zero, prevents a single stuck or flipped bit on an as-yet-unprovisioned chip from causing the chip to fail to be recognised as unprovisioned and thereby enter the failure-response state;

wherein corruption affecting fewer than a failure-threshold number of storage groups is absorbed below a quorum-failure boundary without protocol-level intervention; and

wherein no external signal sequence at the chip's external interface alters, replaces, or extracts the permanent identifier from the storage groups.

10. A system, comprising an integrated-circuit chip configured to implement the methods of claims 1 thru 9, the chip comprising:

a noise-capture circuit configured to sample transient physical phenomena present at and within the chip's silicon at the moment of triggering events, the noise-capture circuit comprising any one or more of: a ring oscillator sampler, a transition-effect ring oscillator, a subthreshold-inverter shot-noise sampler, a metastable arbitration circuit, and a thermal-noise amplifier;

a cryptographic conditioner circuit configured to condition the physical entropy into the chip's permanent identifier and the chip's per-session secret identifier;

a write-once storage array configured to store the chip's permanent identifier according to the method of claim 9;

a finite-state machine configured to implement the provisioning ceremony of claim 1, the attestation method of claim 2, the handshake method of claim 6, the identity dichotomy of claim 7, the failure-response method of claim 8, and the storage method of claim 9;

a wire driver circuit configured to implement the line code of claim 3 and the framing of

claim 4, the wire driver circuit being gated by the failure-response mechanism of claim 8;

exactly one bidirectional input/output pin constituting the chip's only external communication interface with a system in which the chip is installed; and

a free-running ring oscillator internal to the chip and serving as the chip's only clock source;

wherein the chip has no external clock input, no host-authenticated destruct command, no firmware update mechanism, no debug interface, and no in-circuit identity-replacement mechanism;

wherein the chip's finite-state machine transitions are forward-only and enforced by silicon physics rather than by software.

11. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors of a host endpoint communicating with the chip of claim 10, cause the host endpoint to perform:

encoding outgoing host-side bits and decoding incoming chip-side bits according to the line code of claim 3, with the host endpoint configured to use an idle bias value of binary 0;

framing outgoing messages and parsing incoming messages according to the framing convention of claim 4;

constructing outgoing authenticated messages and verifying incoming authenticated messages according to the message format of claim 5;

issuing handshake bits to trigger the entropy-request behaviour of claim 6 and consuming the chip's entropy response;

issuing attestation queries with application-supplied domain identifiers according to claim 2 and verifying the chip's attestation responses against a previously-established public commitment derived from the chip; and

detecting the chip's failure-response state of claim 8 by observing absence of

expected wire heartbeat or absence of expected message response, and treating the failure-response state as indistinguishable from the chip's physical absence for application-level purposes.

12. The method of claim 1, wherein the one or more noise-capture circuits comprise any one or more of: a plurality of free-running ring oscillators with statistical-test entropy extraction; one or more transition-effect ring oscillators; one or more subthreshold-inverter shot-noise sources; one or more metastable arbitration circuits; and one or more thermal-noise amplifier circuits.

13. The method of claim 1, wherein the one or more cryptographic conditioners comprise a cascade of: a chaotic-dynamical amplifier comprising a fixed-point iterated map over a plurality of state buckets, wherein each round applies to each bucket one operation selected from a menu of arithmetic-logic operations by an index derived from the bucket's own current value, a subset of the operations in the menu being information-lossy; followed by a BLAKE3 cryptographic hash function computed over the amplifier's output concatenated with the gathered entropy; such that recovery of the conditioner's input from the conditioner's output requires defeating both the amplifier's path-explosion and information-loss properties and the hash function's preimage resistance, the two being independent security arguments.

14. The method of claim 1, wherein the one or more write-once storage cells comprise any one of: electrically-programmable fuses; antifuses; mask-programmed read-only memory cells; floating-gate cells; write-once magnetic cells; and polymer write-once cells.

15. The method of claim 7, further comprising: receiving, at the chip within an established authenticated session, a state-query message; and responding with an authenticated in-session message whose protected payload comprises a packed state vector representing internal chip state, the state vector including a storage-integrity classification, and in some embodiments further including a session counter, a lifetime-corruption counter, and a free-running boot-epoch timestamp; wherein the

response is of the same wire format, the same payload size, and the same authenticator construction as every other in-session message of the protocol, such that the semantic category of the exchange is not determinable by observation of the wire; and wherein the response is cryptographically bound to the session’s chain state, such that decoding the response against any divergent chain state yields output indistinguishable from a corrupted message and reveals neither the state vector nor any identity of the chip.

16. The method of claim 15, wherein the boot-epoch timestamp is expressed in oscillation counts of the hydrogen-1 hyperfine transition according to the temporal framework described in the related TOKEN application identified in the cross-reference section of the present specification.

17. The method of claim 3, wherein the single electrical conductor comprises any one of: an electrical single-conductor with external pull-down; an optical fiber with optical bias; an inductively-coupled link; a capacitively-coupled link; a radio-frequency link with biased idle state; and an acoustic link.

18. The method of claim 3, wherein a bit rate of the line code is established by the chip’s free-running ring oscillator and is recovered by the host endpoint by means of the automatic clock-recovery property of the line code, such that the chip and the host endpoint do not share an external clock reference.

19. The method of claim 3, wherein the external pull-down resistor is replaced by an external pull-up resistor, the wire’s idle state is at a high logic level, both endpoints release the wire during the other endpoint’s slot, and both endpoints invert the idle bias values relative to the method of claim 3, thereby providing an alternative analog electrical configuration without changing the protocol semantics.

20. The method of claim 5, wherein the keyed cryptographic hash comprises a BLAKE3 keyed hash, the first and second session pads are derived as BLAKE3 hashes of the session-shared secret and of its bitwise complement respectively, the session ratchet advances each pad to the BLAKE3 hash of the payload value of the present exchange,

the masked-commitment field is 256 bits equal to the BLAKE3 output width, and the masked-message field is 256 bits in the case of a 256-bit payload value.

21. The method of claim 5, wherein, in some embodiments for deployment over a communication medium outside the chip’s physical trust boundary, the session-shared secret from which the session pads are derived is itself established by a key-agreement step layered beneath the masked-field construction, the key-agreement step comprising any one of: an elliptic-curve Diffie-Hellman exchange; a post-quantum key-encapsulation mechanism; and a pre-shared-key arrangement; the masked-field construction of claim 5 being unchanged thereby.

22. The method of claim 7, wherein the cryptographic invariant comprises a comparison of a verification field in the authenticated message against two candidate expected values, one expected value being a keyed hash of a recovered message under session pads of the chip and the other expected value being a bitwise complement of the keyed hash, the polarity of the matching expected value determining the selection of the active commitment key between the permanent identifier and the per-session secret identifier.

23. The method of claim 8, wherein the drive-enable signal is computed as the logical AND of a chip-operational status signal derived combinationally from the byzantine quorum check of claim 9, a slot ownership indicator of the endpoint, and a logical wire output value of the endpoint.

24. The method of claim 9, wherein the configured minimum number of valid storage groups for the byzantine quorum check is at least two valid storage groups out of a total of four storage groups, with the recovered identifier required to be in cryptographic agreement across all storage groups satisfying the validity check.

25. The method of claim 9, wherein the validity check applied to each storage group requires that a bitwise exclusive-or of the cells in the storage group equals an all-ones value, and that neither cell in the storage group is at an all-zero state.

26. The system of claim 10, wherein the chip is fabricated in a complementary metal-oxide semiconductor process technology with a feature size of any one of: 350 nanometer, 180 nanometer, 130 nanometer, 90 nanometer, 65 nanometer, 40 nanometer, 28 nanometer, or smaller; and wherein the chip's die area excluding pads is between approximately 0.1 square millimeters and approximately 4.0 square millimeters.

27. The system of claim 10, wherein the chip provides device-identity attestation services to one or more downstream applications selected from: Internet-of-Things device identity without manufacturer involvement after initial provisioning; hardware-wallet seed storage with no factory-injected keys; user-sovereign identity binding as one input to a device-binding function described in the related TOKEN application identified in the cross-reference section of the present specification; automotive secure-boot identity surviving ownership transfer without re-provisioning; medical device patient-private binding; industrial control attestation in environments skeptical of vendor trust; and defense supply-chain post-fabrication substitution detection.

28. The non-transitory computer-readable medium of claim 11, wherein the host endpoint is implemented in any one of: a microcontroller communicating with the chip over a single-channel bus; a general-purpose computer with a single-channel interface adapter; a field-programmable gate array with a soft single-channel driver; an application-specific integrated circuit with a hardened single-channel interface; and a mobile device with a single-channel near-field interface; and wherein the host endpoint is configured to simultaneously serve a plurality of applications that hold independent domain identifiers and that produce independent attestations from a single chip.

29. The system of claim 10, further comprising one or more anti-invasive features selected from: a top-metal active mesh covering the write-once storage cells and the cryptographic conditioner, the mesh configured to detect penetration and to cause the chip's finite-state machine to enter the failure-response state of claim 8; one or more passive shielding layers comprising metallization patterns configured to obstruct

laminographic X-ray imaging of the chip's internal structure; an irradiation-detection circuit configured to cause the chip's finite-state machine to enter the failure-response state of claim 8 upon detection of an attempt at invasive analysis using charged-particle or photon-based probes; one or more dummy circuit blocks resembling but not implementing the cryptographic functions of the chip; and a tamper-evident package coating configured to render decapping observable.

30. The system of claim 10, further comprising one or more operating-envelope sensors and a monitor configured to hold the chip's finite-state machine inoperative and to gate off the chip's wire driver, rendering the chip's wire behaviour indistinguishable from that of an unpowered or absent chip, whenever any condition monitored by the sensors is outside a configured valid range, the monitored conditions comprising any one or more of: a supply voltage below an undervoltage threshold or above an overvoltage threshold; a rate of change of supply voltage exceeding a threshold; a die temperature below a low-temperature bound or above a high-temperature bound; a rate of change of die temperature exceeding a threshold; an internal clock frequency derived from the chip's free-running ring oscillator outside a valid band; an ambient electromagnetic field strength exceeding a threshold; an incident light or charged-particle flux exceeding a threshold; a static or low-frequency magnetic field or magnetic anomaly outside a valid range; a mechanical shock, vibration, or acceleration exceeding a threshold; an ambient pressure outside a valid range; and a chemical signature associated with chemical decapsulation; wherein the out-of-range condition is latched and is cleared only by a complete power-on reset recognised by the chip's supply falling below a hold-off voltage sufficient to guarantee discharge of the chip's volatile state, so that operation resumes only after all monitored conditions are within their valid ranges and a genuine power cycle has occurred, never by a partial recovery while a condition transiently returns within range; and wherein the burn-initiation energy-gate of claim 32 is a burn-specific application of the undervoltage condition and the operating-envelope monitor extends the same principle to all operation of the chip.

31. The method of claim 3, wherein a receiving endpoint of the line code samples the

channel state at a sampling rate substantially higher than the slot rate and accumulates the samples across the slot width by means of an up/down counter, the counter being incremented on each sample at the channel’s high logic level and decremented on each sample at the channel’s low logic level, and wherein the receiving endpoint decodes each slot’s bit value as the sign of the accumulated counter at the slot boundary, said sign-of-sum decode being mathematically equivalent to a binary majority vote across the samples of the slot and therefore to a median filter of width equal to the number of samples per slot, such that channel noise events of duration shorter than one half of the slot width are filtered out without bit error irrespective of noise event amplitude.

32. The method of claim 1, wherein the writing of the permanent identifier to the chip’s write-once storage cells is gated by an energy-budgeted burn-initiation check comprising:

storing electrical energy in an energy storage element coupled to the chip;

measuring, by a sensor, the energy stored in the energy storage element against a configured threshold sufficient to drive the entire writing of the permanent identifier to completion at a burn voltage required by the chip’s write-once storage cells;

withholding initiation of the writing until the sensor reports the configured threshold is met; and

upon initiation of the writing, electrically isolating the energy storage element from the chip’s external power supply by a switching element coupled between them, such that the writing proceeds to completion under power supplied by the energy storage element and is independent of the chip’s external power supply for the duration of the writing;

wherein interruption of the chip’s external power supply at any time before the sensor reports the configured threshold met leaves no write-once storage cell written and the chip is observed at next reset as being in the unprovisioned KORE state, the chip’s owner being capable of re-attempting the provisioning method of claim 1 with fresh entropy drawn at the moment of the re-attempt without any operator-side recovery

procedure being required; and

wherein the entropy-gathering and cryptographic-conditioning sub-steps preceding the writing operate on volatile registers internal to the chip and write no write-once storage cell, such that interruption of the chip's external power supply during said sub-steps drops the volatile registers and the chip is observed at next reset as being in the unprovisioned KORE state.

33. The method of claim 32, wherein the energy storage element comprises any one or more of: an intrinsic on-die power-supply decoupling capacitance of the chip; an on-package decoupling capacitance coupled to the chip's supply pin; and an external capacitor coupled to the chip's supply pin or to a dedicated burn-rail pin of the chip; and wherein the sensor comprises an analog comparator measuring a terminal voltage of the energy storage element against a reference voltage and emitting a single binary burn-permission signal to the chip's finite-state machine.

34. The method of claim 32, wherein the chip's finite-state machine presents, at every instant during and after the writing of the permanent identifier, exactly one of: (i) the KORE state, in which the burn-initiation check has not been passed and no write-once storage cell has been written; or (ii) the ORA state, in which the burn-initiation check has been passed and the writing has completed; no intermediate state of the chip's finite-state machine between the KORE state and the ORA state being reachable by interruption of the chip's external power supply at any time during the writing, the atomic transition between KORE and ORA being enforced by the energy budget of the energy storage element rather than by firmware, host-endpoint protocol, or operator-side recovery procedure.

35. A method of hardware-rooted cryptographic erasure, comprising:

deriving, at an integrated-circuit chip having a permanent identifier stored in write-once storage cells of the chip and a per-session secret identifier held in volatile registers of the chip and established once per power-on cycle from physical entropy sampled

by noise-capture circuitry resident on the chip's silicon, a content key as a function of the per-session secret identifier, an identifier of a protected payload, and an application-supplied freshness value, the deriving being performed by a cascaded conditioner comprising a chaotic-dynamical amplifier followed by a keyed cryptographic hash function;

retaining the content key only in volatile registers of the chip, the chip having no retention element for the per-session secret identifier or any value derived from the per-session secret identifier other than the volatile registers, and having no path in the chip's mask design by which the per-session secret identifier or any value derived from it can persist across an interruption of the chip's external power supply;

losing, upon any interruption of the chip's external power supply for any duration sufficient to discharge the volatile registers, the per-session secret identifier and the derived content key irrecoverably, the loss being a physical consequence of the discharge of the volatile registers rather than a software-asserted property of the chip's firmware;

establishing, after the chip is restored to operation following the interruption, a new per-session secret identifier from a fresh sample of physical entropy sampled by the noise-capture circuitry, the new per-session secret identifier being cryptographically independent of the prior per-session secret identifier; and

proving the erasure to a verifying application by transmitting, in response to queries from the verifying application, (i) a keyed cryptographic commitment attesting that the chip's permanent identifier is unchanged from the chip's permanent identifier at a prior registration of the chip with the verifying application, and (ii) a session-continuity response that differs from a corresponding response previously recorded by the verifying application from a prior session of the chip, the combination of identity-match with session-mismatch constituting cryptographic proof that the chip's session-state has been lost between the prior session and the current session.

36. The method of claim 35, wherein no protocol-level command, no fresh-session

handshake, and no other in-session signal at the chip's external interface removes the per-session secret identifier or any value derived therefrom; and wherein the only mechanism by which the prior per-session secret identifier can be lost is the physical discharge of the volatile registers that hold it, such that the erasure is enforced by the physical absence of any retention path in the chip's mask design and is not subject to firmware override, host-endpoint instruction, or operator-side recovery procedure.

37. The method of claim 35, wherein verification by the verifying application of the chip's hardware authenticity at the prior registration is established by reliance on an audited manufacturing process of the chip rather than by reliance on a vendor-signed endorsement of the chip, the chip's mask design and fabrication process being subject to public publication and third-party audit, and verification of any specific chip's hardware authenticity reducing to verification of the chip's provenance thru the audited supply chain; the trust anchor of the erasure proof being the audited supply chain and admitting no centralized cryptographic root subject to coercion of a single party.

38. The method of claim 35, wherein the protected payload comprises one or more of: digital content under a licensing arrangement in which a licensor verifies that a licensee's playback device has irreversibly lost the capability to decrypt licensed content upon end-of-rental; digital library media in which a return event is verified cryptographically rather than asserted by client software; a transferable capability between devices in which a source device's loss of the capability is verified before a destination device's grant of the capability is honoured; tamper-evident continuous-operation state of a high-assurance system in which an unexpected session-state loss evidences interruption; and any analogous protected payload in which a remote party requires cryptographic confirmation that a counterparty's previously-held capability is no longer recoverable.

39. The method of claim 35, wherein the verifying application's session-continuity response is generated by the chip as a deterministic function of the per-session secret identifier and a query supplied by the verifying application, such that a subsequent

identical query during the same power-on cycle of the chip returns the identical response, and an identical query following any interruption of the chip's external power supply returns a different response, the identical-return case providing cryptographic proof of session continuity and the different-return case providing cryptographic proof of session-state loss.

40. The method of claim 35, wherein the chip provides the erasure proof without revealing the chip's permanent identifier in plaintext or in any reversible encoding on the chip's external interface, the keyed cryptographic commitment of the identity-match prong being preimage-resistant against recovery of the permanent identifier; and wherein the chip provides the erasure proof without requiring trust in any third party including the chip's manufacturer beyond the manufacturer's role in producing genuine hardware of the published mask design, and without requiring participation of any trusted execution environment attestation chain.

41. The method of claim 7, further comprising proving continuity of the chip's session to a verifying application, by: receiving, at the chip, an identical deterministic query at two distinct times within one power-on cycle of the chip, the chip's response to the query being a deterministic function of the per-session secret identifier and the query; and returning an identical response at both times; wherein the identical response at the second time constitutes cryptographic proof to the verifying application that the chip's volatile state has been continuously preserved between the two times, no interruption of the chip's power supply sufficient to discharge the chip's volatile registers having occurred and no replacement of the chip with a different physical chip having occurred; and wherein a differing response to the identical query constitutes cryptographic proof that the chip's session has been broken by power interruption or physical replacement.

42. The method of claim 4, wherein, in an alternative framing convention, leading binary 0 bits of the payload are not actively driven onto the wire and a single trailing binary 1 bit is transmitted as the message-end anchor, the receiving endpoint reconstructing the suppressed leading zeros by counting slots backward from the trailing bit against the

known message length, the framing cost being exactly one channel slot per message and the framing savings being the count of leading binary 0 bits of the payload.

43. The method of claim 7, wherein the per-session secret identifier is held in the volatile registers as a plurality of register groups, each register group holding the per-session secret identifier in one register and a bitwise complement of the per-session secret identifier in a paired register, each register group verified at use by a validity check requiring that a bitwise exclusive-or of the paired registers equal an all-ones value; and wherein, upon failure of the validity check during a power-on cycle, the chip ceases responding at the chip’s external interface for the remainder of the power-on cycle, recovery comprising interruption of the chip’s power supply followed by a fresh handshake according to the method of claim 6, the chip performing no silent re-derivation of the per-session secret identifier, whereby corruption of the volatile registers by single-event upset is converted into a detectable session break rather than an undetectable divergence of session state.

8 ABSTRACT OF THE DISCLOSURE

A method and apparatus for vendor-free domain-separated cryptographic device attestation. An integrated-circuit chip generates a permanent identifier from physical entropy sampled by on-chip noise-capture circuitry at the moment of an owner-initiated provisioning ceremony, conditions it thru an on-chip cascade of a chaotic-dynamical amplifier and a cryptographic hash, and atomically burns it to write-once storage under an energy-budgeted commit gate. No third party witnesses or signs the provisioning, and the identifier never leaves the chip. Attestations are keyed commitments over an application-supplied domain identifier, cryptographically independent across applications, preventing structural cross-application correlation. A volatile per-session secret, established once per power-on cycle and irrecoverably lost on power interruption, provides session-continuity proof and hardware-rooted provable erasure of session-bound capabilities. The chip's only interface is a single bidirectional pin carrying a mirror-XOR line code; identity storage uses inverted-pair write-once cells with byzantine quorum recovery; every failure mode presents wire silence indistinguishable from physical absence.

9 DRAWINGS

FIG. 1

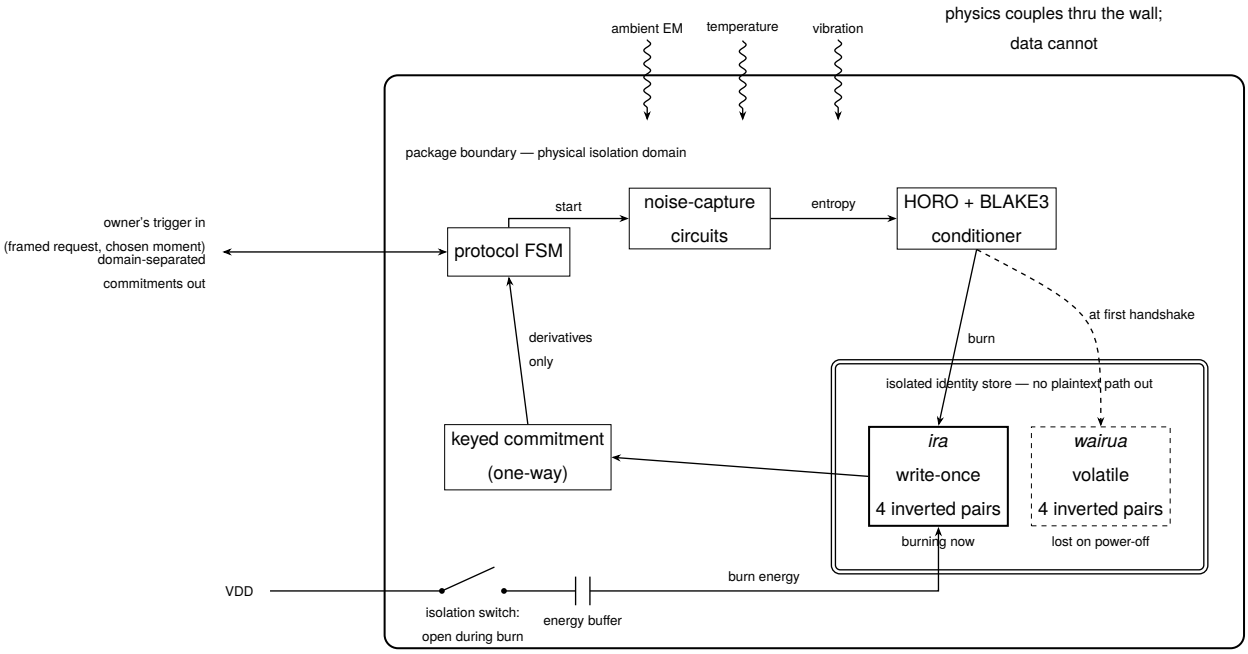
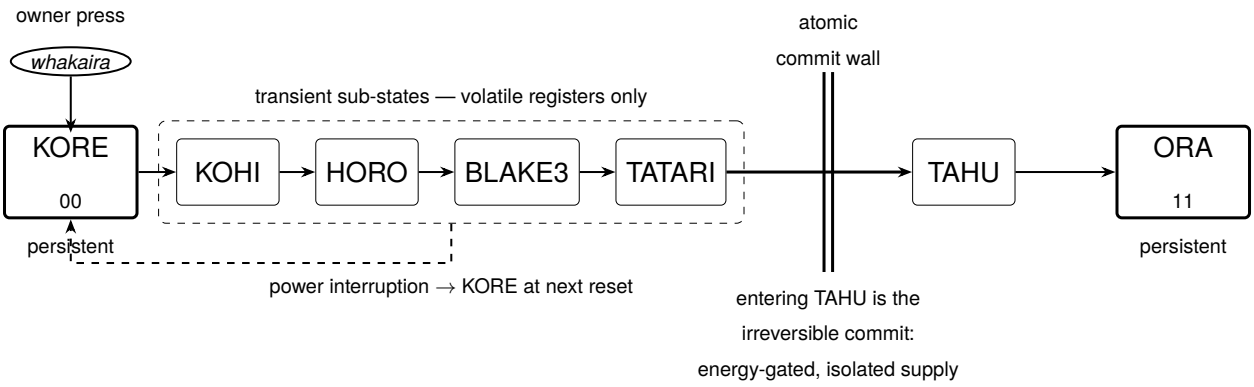


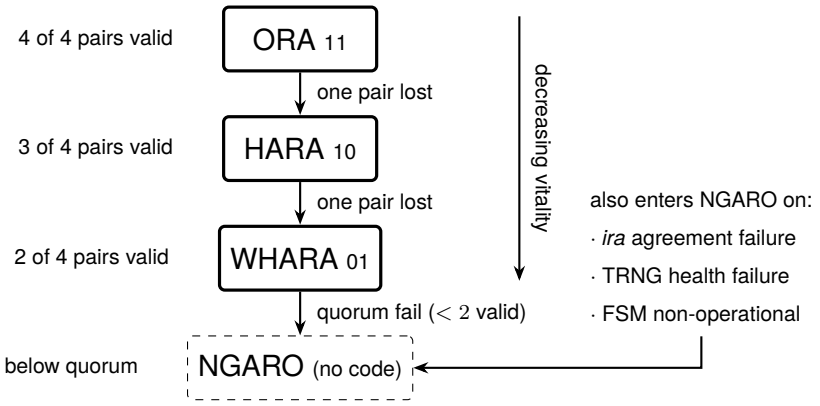
FIG. 2

The provisioning lifecycle: the chip's birth.



Exactly two states ever persist across a reset during provisioning: KORE before commit and ORA after. The dashed sub-states (KOHI thru TATARI) run on volatile registers only and leave no trace if interrupted. Entering TAHU is the single irreversible crossing of the commit wall: once the energy-gated burn begins, the energy buffer is isolated and the burn runs to completion regardless of any external power interruption, so TAHU lies on the committed side of the wall.

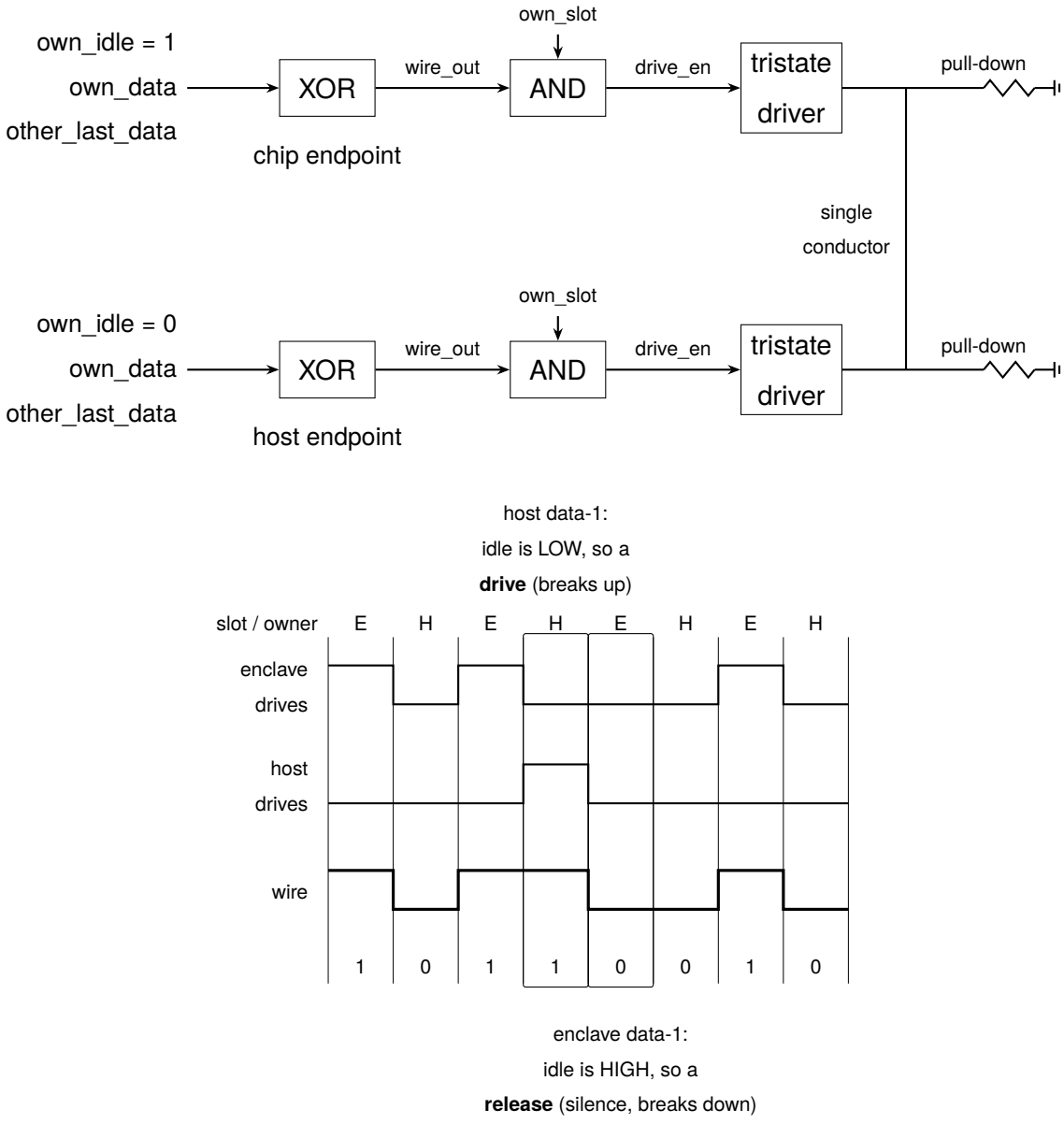
The operational degradation ladder: the chip's aging.



silent zone: no wire output; indistinguishable from an absent chip

After provisioning, corruption of the inverted-pair storage degrades the chip down the vitality ladder, the two-bit code equalling the count of valid pairs. Below byzantine quorum, or on any verify / TRNG / FSM failure, the chip enters NGARO and goes silent. NGARO is recoverable only by power-on reset, and only once the underlying failure has cleared.

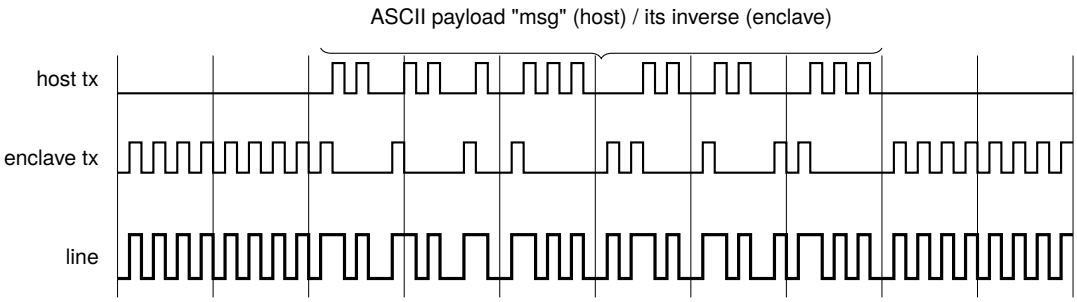
FIG. 3



Two minimal signalling events. Because the endpoints hold opposite idle biases, the same data bit is signalled by opposite physical gestures: a one from the host (idle LOW) is an active *drive* that breaks the line upward, while a one from the enclave (idle HIGH) is a *release* that lets the line fall to its pull-down, a silence that breaks the line downward. Every slot has a single defined source, so the two endpoints never contend. FIG. 4 carries this mechanism over a full 80-slot exchange on a real payload, where the composite conductor settles to an exact 40/40 direct-current balance.

FIG. 4

A full 80-slot full-duplex exchange carrying the ASCII payload “msg” from the host, with the enclave transmitting concurrently.



host tx: 15 HIGH / 65 LOW
enclave tx: 25 HIGH / 55 LOW
line (their XOR): 40 HIGH / 40 LOW

Each endpoint’s stream is individually unbalanced; the line, their mirror-XOR superposition, is exactly DC-balanced. The enclave stream is the bitwise inverse of the host’s (idle 1 vs. idle 0).

FIG. 5

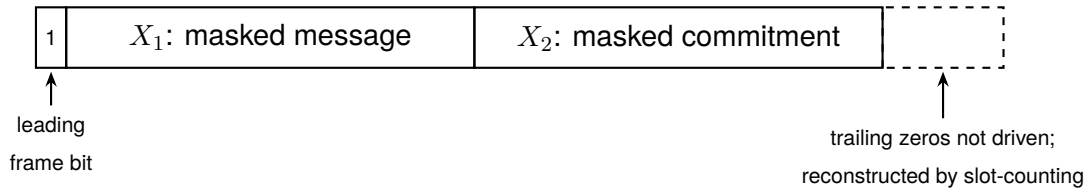

1
X_1: masked message
X_2: masked commitment
leading
frame bit
trailing zeros not driven;
reconstructed by slot-counting

FIG. 6

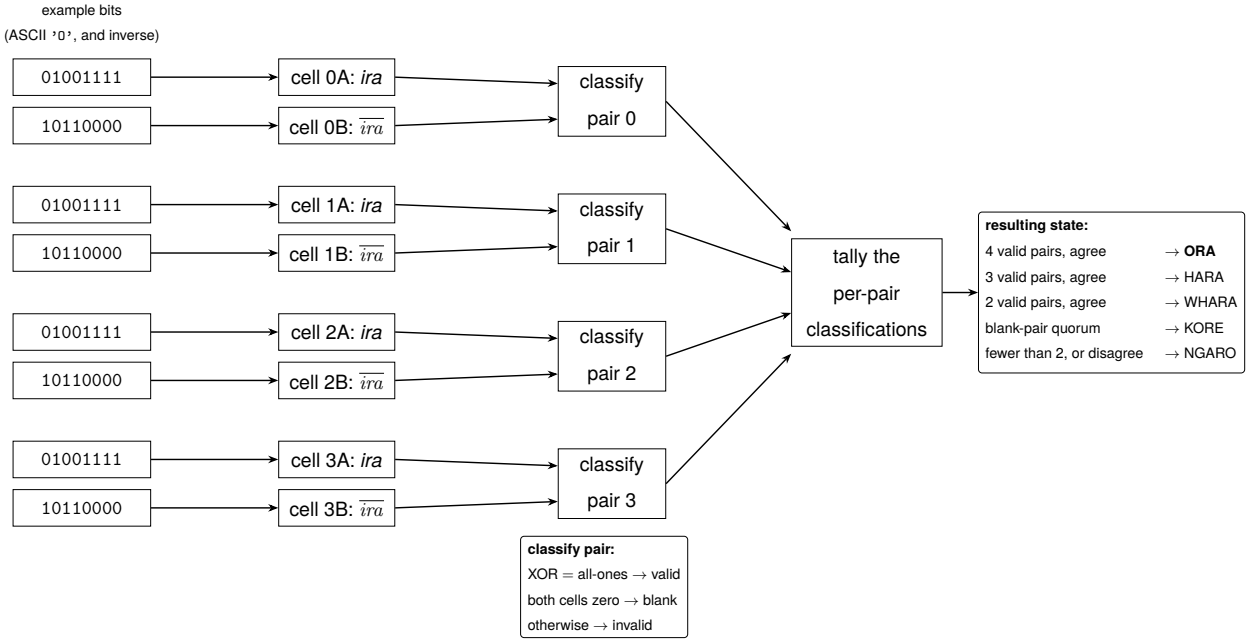
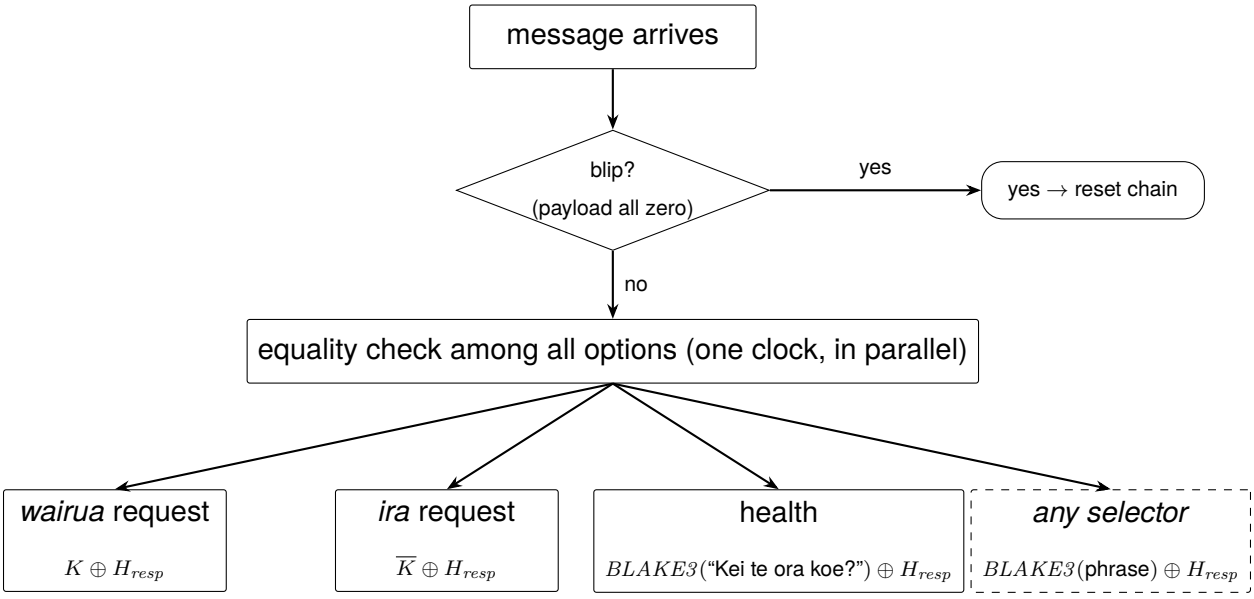

example bits
(ASCII 'O', and inverse)
01001111
10110000
cell 0A: ira
cell 0B: $\overline{ira}$
classify pair 0
cell 1A: ira
cell 1B: $\overline{ira}$
classify pair 1
cell 2A: ira
cell 2B: $\overline{ira}$
classify pair 2
cell 3A: ira
cell 3B: $\overline{ira}$
classify pair 3
classify pair:
XOR = all-ones → valid
both cells zero → blank
otherwise → invalid
tally the per-pair classifications
resulting state:
4 valid pairs, agree → ORA
3 valid pairs, agree → HARA
2 valid pairs, agree → WHARA
blank-pair quorum → KORE
fewer than 2, or disagree → NGARO

FIG. 7


message arrives
blip?
(payload all zero)
yes
yes → reset chain
no
equality check among all options (one clock, in parallel)
wairua request
$K \oplus H_{resp}$
ira request
$\overline{K} \oplus H_{resp}$
health
$BLAKE3$(“Kei te ora koe?”) $\oplus H_{resp}$
any selector
$BLAKE3$(phrase) $\oplus H_{resp}$